Generalized AKM: Theory and Evidence*

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Abstract

We develop a semiparametric leave-one-out estimator for variance components in two-way fixed-effects models where covariates enter through an unknown function approximated by a growing series basis. We prove consistency and asymptotic normality under heteroskedasticity and many fixed effects, identifying a smoothness condition that is stronger than what linear functionals require but mild in wage applications. In Portuguese matched employer–employee data, standard linear AKM decompositions become unstable when firm-level controls are added, with variance components inflating by an order of magnitude and sorting reversing sign. Estimating a sequence of increasingly flexible specifications shows that the instability is driven by nonlinearity, not slope heterogeneity: a flexible polynomial basis eliminates most of the distortion.

Keywords: wage decomposition, many regressors, semiparametric model, series estimator

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1. Introduction

Decomposing wage dispersion into worker and firm components is a basic question in labor economics. Since Abowd, Kramarz, and Margolis (1999), the standard approach has been to estimate a two-way fixed-effects model—commonly known as AKM—in matched employer–employee data and report how much of the overall wage variation is attributable to workers, to firms, and to their sorting. A large literature has refined the statistical foundations of this decomposition, most notably through the leave-out bias correction of Kline, Saggio, and Sølvsten (2020), which delivers valid inference under heteroskedasticity and many fixed effects. Throughout, however, both the original estimator and its modern refinements assume that observed covariates—age, education, occupation, and increasingly firm-side variables such as revenue or assets—enter log wages through a known linear function. This is hard to reconcile with evidence that returns to experience and schooling are nonlinear (Mincer 1974; Card et al. 2018), and there is no strong reason to expect firm-level controls to be additively separable from worker characteristics. Whether and how this linearity restriction distorts the decomposition in practice is an open question.

This paper develops a semiparametric estimator for variance components in models where covariates enter through an unknown function. Correctly attributing wage variation to workers, firms, and their sorting requires isolating the role of observed characteristics from unobserved heterogeneity; if the functional form linking covariates to wages is misspecified, the decomposition inherits the distortion. The model retains the two-way fixed-effects structure of AKM but replaces the linear control function with an unspecified smooth function, approximated by a growing polynomial or spline basis. Our estimator combines the leave-one-out bias correction of Kline, Saggio, and Sølvsten (2020) with this flexible approximation: the standard plug-in decomposition is corrected using leave-one-out residual variances computed from the augmented regressor set that includes both the high-dimensional fixed effects and the sieve basis. We show that this estimator is consistent and asymptotically normal for the original variance-component targets under heteroskedasticity, a growing number of fixed effects, and an explicit smoothness condition on the unknown function.

The main theoretical challenge is that three problems must be handled jointly: many-regressor bias from high-dimensional fixed effects, heteroskedasticity-robust variance estimation, and approximation error from the unknown control function. Existing leave-out estimators solve the first two but assume linearity; existing semiparametric series theory handles the third but targets linear functionals of the coefficients, not quadratic forms. Because variance components are quadratic in the coefficient vector, approximation error enters both the plug-in estimator and the leave-out bias correction, and the two sources must vanish jointly. This imposes a smoothness requirement on the unknown function that is stronger than what standard semiparametric estimation of regression coefficients demands: the function must have more continuous derivatives than the number of continuous covariates entering flexibly. In wage applications, where those

covariates are quantities such as age, firm revenue, or assets, the condition is mild. We prove consistency, a chi-squared limit when the variance-component target has fixed rank, and a normal limit for the empirically relevant case where the rank grows with sample size. A random-projection approximation extends the estimator to the large-scale administrative datasets typical of this literature.

The simulation study confirms these mechanisms. Under strong nonlinearity, richer polynomial bases substantially reduce bias, especially when misspecification interacts with near-collinear regressors or many fixed effects. When the true relationship is linear or mildly nonlinear, all estimators perform similarly, so the additional flexibility imposes little cost when it is not needed. The simulations also show that finite-sample inference is harder than point estimation: in the most demanding designs, confidence interval coverage falls well below nominal even when bias is modest. For applied work, the estimator is therefore most useful as a decomposition and point-estimation tool.

We apply the estimator to Portuguese linked employer–employee administrative data for the period 2008–2018, estimating a sequence of increasingly flexible specifications within two samples that differ in the richness of observed covariates. When firm-level controls such as revenue and assets are included, the standard linear specification yields an implausible decomposition: worker-effect and firm-effect variances inflate by an order of magnitude and the sorting term reverses sign. The source of the instability is functional-form misspecification. Revenue and assets vary predominantly across firms rather than within firms over time; forcing them through a linear model alongside firm fixed effects creates collinearity that the bias correction amplifies. Allowing these controls to enter nonlinearly eliminates most of the distortion and returns the decomposition to plausible magnitudes; heterogeneous slopes across demographic groups provide a further but second-order refinement. In the parsimonious sample with only gender and age, the decomposition is stable throughout. Part of what appears as worker–firm sorting or inflated firm premia in linear AKM thus reflects misspecified returns to observed characteristics, not a genuine correlation between unobserved worker and firm types.

Contribution to the literature. The paper contributes to three strands of the literature. First, it extends the leave-out variance-component framework of Kline, Saggio, and Sølvsten (2020) from linear many-regressor models to semiparametric models with a growing approximation space, providing the conditions under which leave-out validity survives flexible approximation of the control function. Second, it complements the semiparametric series literature (Donald and Newey 1994; Cattaneo, Jansson, and Newey 2018a, 2018b) by moving from slope-type targets to variance components, identifying the stronger smoothness condition this shift requires. Third, it contributes to the empirical literature on the sensitivity of AKM decompositions (Andrews et al. 2008; Bonhomme et al. 2023; Bonhomme, Lamadon, and Manresa 2019) by showing that much of the instability documented when adding controls to linear AKM can be attributed to functional-form misspecification rather than to a genuine change in the underlying economic structure. Our approach differs from grouped-firm or latent-type models (Bonhomme, Lamadon,

and Manresa 2019) in that it preserves the original variance-component targets and asks whether leave-out estimation remains valid once the control function is flexibly specified, rather than replacing the AKM structure with a different earnings model.

Outline. The remainder of the paper is organized as follows. Section 2 introduces the semiparametric model and formalizes the variance component targets. Section 3 derives the estimator and its finite-sample representation. Section 4 establishes consistency, summarizes the strengthened smoothness condition, and collects the core rate restrictions used throughout the theory. Section 5 presents the random projection approximation. Section 6 derives the fixed-rank and growing-rank limiting distributions of our proposed estimator. Section 7 reports our simulation exercise, while Section 8 illustrates an empirical application on linked employer-employee data. Section 9 concludes.

2. Setup

We consider the following semiparametric model,

$$y_i = x_i'\beta + f(z_i) + e_i, \quad i = 1, \ldots, n, \tag{1}$$

where the regressors $x_i \in \mathbb{R}^p$, $z_i \in \mathbb{R}^d$ are non-random. The unknown function $f(z)$ belongs to the class of smooth and real-valued functions, $f \in \mathcal{F}$. The unobserved errors $\{e_i\}_{i=1}^n$ are mutually independent and obey $\mathbb{E}[e_i] = 0$, but may possess observation-specific variances $\mathbb{E}[e_i^2] = \sigma_i^2$.

Our object of interest is a quadratic form $\theta := \beta' A \beta$ for some known non-random symmetric matrix $A \in \mathbb{R}^{p \times p}$ of rank r. This is a quantity of interest in a wide range of economic applications: two key examples are summarized next.

Example 1 (Generalized analysis of variance)**.** Consider observations arranged into N groups with T_g observations in the g-th group. Since Fisher (1928), the standard analysis of variance model assumes:

$$y_{gt} = \alpha_g + z_{gt}'\delta + \varepsilon_{gt}, \quad g = 1, \ldots, N, \quad t = 1, \ldots, T_g,$$

where α_g are group-specific fixed effects, and z_{gt} is a vector of exogenous covariates. The focus here is the variability in the outcome variable attributable to groups, or

$$\sigma_\alpha^2 = \frac{1}{n} \sum_{g=1}^{N} T_g \left(\alpha_g - \bar{\alpha}\right)^2$$

with $n := \sum_{g=1}^N T_g$, and $\bar{\alpha} := n^{-1} \sum_{g=1}^N T_g \alpha_g$. Our approach relaxes the linearity assumption on covariates and instead assumes any form of unspecified functional dependence, that is

$$y_{gt} = \alpha_g + f(z_{gt}) + \varepsilon_{gt}, \quad f \in \mathcal{F}, \quad g = 1, \ldots, N, \quad t = 1, \ldots, T_g.$$

We can represent it as in (1), defining $i := i(g,t)$ with $i(\cdot,\cdot)$ being a bijective function, $y_i := y_{gt}$, $z_i := z_{gt}$ and $e_i := \varepsilon_{gt}$,

$$x_i := d_i, \quad \beta := (\alpha_1, \ldots, \alpha_N)', \quad d_i := (\mathbb{1}\{g=1\}, \ldots, \mathbb{1}\{g=N\})'.$$

Our object of interest σ_α^2 can be represented here as $\beta' A \beta$, with

$$A := \begin{pmatrix} A_d' A_d & 0 \\ 0 & 0 \end{pmatrix}, \quad A_d := \frac{1}{\sqrt{n}}(d_1 - \bar{d}, \ldots, d_n - \bar{d}), \quad \bar{d} := \frac{1}{n}\sum_{i=1}^{n} d_i.$$

Example 2 (Generalized AKM)**.** Our second and leading example is a classic wage decomposition model proposed in Abowd, Kramarz, and Margolis (1999). It models the log wage determination as an additive function of worker fixed effects, firm fixed effects, and a linear function of strictly exogenous covariates. Specifically,

$$y_{gt} = \alpha_g + \psi_{j(g,t)} + z_{gt}'\delta + \varepsilon_{gt}, \quad g = 1, \ldots, N, \quad t = 1, \ldots, T_g.$$

Here, α_g and $\psi_{j(g,t)}$ capture the gth worker and jth firm unobserved heterogeneity component respectively, and z_{gt} is a vector of exogenous regressors. The bijective function $j(\cdot,\cdot) : \{1, \ldots, N\} \times \{1, \ldots, \max_g T_g\} \to \{0, \ldots, J\}$ maps $n = \sum_{g=1}^{N} T_g$ person-year observations to one of $J+1$ firms. One of the model's objectives is to quantify how much of the variability in log wages is determined by firms,

$$\sigma_\psi^2 = \frac{1}{n}\sum_{g=1}^{N}\sum_{t=1}^{T_g}\left(\psi_{j(g,t)} - \bar{\psi}\right)^2,$$

where $\bar{\psi} = \frac{1}{n}\sum_{g=1}^{N}\sum_{t=1}^{T_g}\psi_{j(g,t)}$. Given evidence that returns to worker characteristics are nonlinear (Mincer 1974; Card et al. 2018), part of the measured variance of worker and firm fixed effects may stem from the linear restriction on z_{gt}. Our methodology replaces it with unspecified functional dependence,

$$y_{gt} = \alpha_g + \psi_{j(g,t)} + f(z_{gt}) + \varepsilon_{gt}, \quad f \in \mathcal{F}, \quad g = 1, \ldots, N, \quad t = 1, \ldots, T_g.$$

This formulation can capture nonlinear interactions between worker-level and firm-level characteristics that vary across observed groups in the population.

Given that the common covariates z_{gt} and the firm assignments $j(\cdot,\cdot)$ obey a strict exogeneity condition, we can rewrite the equation above as in (1) with

$$x_i := (d_i', h_i')', \quad \beta := (\alpha', \psi')', \quad \alpha := (\alpha_1, \ldots, \alpha_N)' + \mathbb{1}_N'\psi_0, \quad \psi = (\psi_1, \ldots, \psi_J)' - \mathbb{1}_J'\psi_0,$$

defining y_i, z_i, and e_i as in Example 1, and $h_i := (\mathbb{1}\{j(g,t)=1\}, \ldots, \mathbb{1}\{j(g,t)=J\})'$. The

parameter of interest σ_ψ^2 then can be rewritten as $\beta' A_\psi \beta$ with

$$A_\psi := \begin{pmatrix} 0 & 0 & 0 \\ 0 & A_h' A_h & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad A_h := \frac{1}{\sqrt{n}}(h_1 - \bar{h}, \ldots, h_n - \bar{h}), \quad \bar{h} := \frac{1}{n}\sum_{i=1}^{n} h_i.$$

2.1. Why existing estimators do not directly apply

Throughout, our object of interest is the quadratic form $\theta = \beta' A \beta$, defined on the high-dimensional fixed-effect coefficients, while the control function $f(z)$ is estimated through a growing approximation space. Several existing estimation strategies address parts of this setting; we review each in turn and explain why none covers the full problem.

KSS-style leave-out estimation. Kline, Saggio, and Sølvsten (2020) provide valid leave-out estimation of the same quadratic target $\theta = \beta' A \beta$ in finite-dimensional linear many-regressor models under arbitrary heteroskedasticity. Our setting preserves that object, but drops the assumption that observables enter through a finite-dimensional linear control. Once $f(z)$ is replaced by a series approximation $p_k(z)'\alpha$ with $k \to \infty$, validity is no longer automatic: both $\hat{\beta}$ and the leave-out variance estimators inherit approximation error, so one must re-establish that leave-out validity survives sieve approximation.[1]

Semiparametric series and orthogonalization. Series estimators in partially linear models (Donald and Newey 1994; Cattaneo, Jansson, and Newey 2018a) and orthogonalized or cross-fit procedures (Chernozhukov et al. 2018; Bonhomme, Jochmans, and Weidner 2025) allow an unknown nuisance function, but they target low-dimensional slope coefficients or other approximately linear functionals. None provides inference for quadratic forms under the high-dimensional fixed-effect structure and heteroskedasticity that define our setting.

3. Finite-sample properties

From now on, we restrict the analysis to model (1). To derive an estimator of β, one must regress y_i on x_i and *functions* of z_i. To this end, let $p^1(z), \ldots, p^k(z)$ be some *approximating* functions, and let $p_k(z) := (p^1(z), \ldots, p^k(z)) \in \mathbb{R}^k$ collect them into a vector. We assume that the class of functions $\mathcal{F}$ to which f belongs can be well approximated by linear combinations of the basis functions $p^1(z), \ldots, p^k(z)$.

Define M_{ij} as a (i,j)th element of $M := I_n - P_k(P_k' P_k)^{-1} P_k'$ with

$$P_k := (p_k(z_1), \ldots, p_k(z_n)) \in \mathbb{R}^{n \times k},$$

1. A separate strand of the literature relaxes the economic structure of AKM itself. For example, Bonhomme, Lamadon, and Manresa (2019) model latent worker and firm types and reduce dimensionality through grouping. These approaches replace the original variance-component target with a different earnings structure, rather than preserving $\theta = \beta' A \beta$ under an unknown nuisance function.

and let the partialed-out design matrix $S_{xx} := \sum_{i=1}^n \sum_{j=1}^n M_{ij} x_i x_j'$ have a full column rank. Then, the estimator of β is defined as

$$\hat{\beta} := S_{xx}^{-1} \sum_{i=1}^n \sum_{j=1}^n M_{ij} x_i y_j.$$

Under some regularity conditions, the bias of $\hat{\beta}$ is negligible for large k. Rewrite it as:

$$\begin{aligned}
\hat{\beta} &= S_{xx}^{-1} \sum_{i=1}^n \sum_{j=1}^n M_{ij} x_i \left(x_j' \beta + f(z_j) + e_j \right) \\
&= S_{xx}^{-1} \sum_{i=1}^n \sum_{j=1}^n M_{ij} x_i x_j' \beta + S_{xx}^{-1} \sum_{i=1}^n \sum_{j=1}^n M_{ij} x_i f(z_j) + S_{xx}^{-1} \sum_{i=1}^n \sum_{j=1}^n M_{ij} x_i e_j \\
&= \beta + S_{xx}^{-1} \sum_{i=1}^n \sum_{j=1}^n M_{ij} x_i f(z_j) + S_{xx}^{-1} \sum_{i=1}^n \sum_{j=1}^n M_{ij} x_i e_j := \beta + B + U.
\end{aligned}$$

Because of the zero-mean assumption on errors, $\mathbb{E}[U] = 0$ holds, and the bias in the estimator is reflected in the B term only. Under some conditions discussed below,

$$B = o(1)$$

as the number of approximating functions goes to infinity, $k \to \infty$.

Analogously to the fully parametric setup examined by Kline, Saggio, and Sølvsten (2020), the plug-in estimator of the quadratic form $\hat{\theta}_{\text{PI}} := \hat{\beta}' A \hat{\beta}$ is biased, because

$$\mathbb{E}[\hat{\theta}_{\text{PI}} - \theta] = \text{trace}(A \, \text{var}[\hat{\beta}]) = \sum_{i=1}^n B_{ii} \sigma_i^2,$$

where $B_{ii} := \sum_{j=1}^n M_{ij} x_j' S_{xx}^{-1} A S_{xx}^{-1} \sum_{j=1}^n M_{ij} x_j$ measures the influence of the ith squared error e_i^2 on $\hat{\theta}_{\text{PI}}$. In cases when $p/n \approx 0$, individual elements B_{ii} are close to zero, and the bias is negligible. However, with increasing p, the bias is more pronounced and needs to be accounted for.

We introduce some notation to describe our estimator. Collect x_i in a matrix as $X := (x_1, \ldots, x_n)' \in \mathbb{R}^{n \times p}$. Then, $W := (X, P_k) \in \mathbb{R}^{n \times (p+k)}$ is a matrix containing the full set of regressors, with individual vectors w_i, and $W = (w_1, \ldots, w_n)'$. Define

$$\hat{\gamma} := \left(\sum_{j=1}^n w_j w_j' \right)^{-1} \sum_{j=1}^n w_j y_j,$$

and the leave-one-out version of it,

$$\hat{\gamma}_{-i} := \left(\sum_{j=1}^n w_j w_j' - w_i w_i' \right)^{-1} \left(\sum_{j=1}^n w_j y_j - w_i y_i \right).$$

Note that the first p elements of $\hat{\gamma}$ are $\hat{\beta}$, and the remaining k elements are basis coefficients

from the function approximation. Similarly to Kline, Saggio, and Sølvsten (2020), we estimate θ as

$$\hat{\theta} := \hat{\beta}' A \hat{\beta} - \sum_{i=1}^{n} B_{ii} \hat{\sigma}_i^2, \tag{2}$$

where $\hat{\sigma}_i^2$ is a leave-one-out estimator of the individual variance of e_i defined as

$$\hat{\sigma}_i^2 := y_i \left(y_i - w_i' \hat{\gamma}_{-i} \right). \tag{3}$$

We note that computing $\hat{\gamma}_{-i}$ for each $i = 1, \ldots, n$ is computationally costly in large-scale applications. To avoid this, we represent (3) as

$$\hat{\sigma}_i^2 = \frac{y_i \hat{e}_i}{M_{W,ii}}, \tag{4}$$

where $M_{W,ii}$ is ith diagonal element of the matrix $M_W := I_n - W(W'W)^{-1}W'$ projecting onto the complement of the column space spanned by x_i and functions of z_i, and $\hat{e}_i := \sum_{j=1}^{n} M_{ij}(y_j - x_j \hat{\beta})$ are residuals.

Our estimator $\hat{\theta}$ is *exactly* unbiased if *(i)* $\hat{\beta}$ is unbiased, *(ii)* $\hat{\sigma}_i^2$ is unbiased for each i. For a moment, suppose that $\mathbb{E}[\hat{\beta}] = \beta$ holds and $f \in \mathcal{F}_{\text{linear}}$: that is, the estimator of the "slope" β is unbiased, and the unknown function $f(z) = z'\alpha$ is linear in z. Under these conditions, we have

$$\begin{aligned}
\mathbb{E}[\hat{\sigma}_i^2] &= \mathbb{E}\left[y_i \left(y_i - x_i' \hat{\beta}_{-i} - p_k(z_i)' \hat{\alpha}_{-i} \right) \right] \\
&= \mathbb{E}\left[\left(x_i' \beta + f(z_i) + e_i \right) \left(x_i' \beta + f(z_i) + e_i - x_i' \hat{\beta}_{-i} - p_k(z_i)' \hat{\alpha}_{-i} \right) \right] \\
&= \mathbb{E}\left[\left(x_i' \beta + f(z_i) + e_i \right) \left(x_i'(\beta - \hat{\beta}_{-i}) + (f(z_i) - p_k(z_i)' \hat{\alpha}_{-i}) + e_i \right) \right] \\
&= \left(x_i' \beta x_i' + f(z_i) x_i' \right) \mathbb{E}[\beta - \hat{\beta}_{-i}] + \left(x_i' \beta + f(z_i) \right) \mathbb{E}[f(z_i) - p_k(z_i)' \hat{\alpha}_{-i}] + \sigma_i^2 \\
&= \sigma_i^2
\end{aligned}$$

where we use the fact that $\mathbb{E}[e_i x_i'(\beta - \hat{\beta}_{-i})] = \mathbb{E}[e_i]\mathbb{E}[x_i'(\beta - \hat{\beta}_{-i})] = 0$, because $\hat{\beta}_{-i}$ is independent of e_i, and similarly $\mathbb{E}[e_i(f(z_i) - p_k(z_i)' \hat{\alpha}_{-i})] = \mathbb{E}[e_i]\mathbb{E}[f(z_i) - p_k(z_i)' \hat{\alpha}_{-i}] = 0$, because $\hat{\alpha}_{-i}$ is independent of e_i.

However, for general classes of functions $\mathcal{F}$, the exact unbiasedness of $\hat{\sigma}_i^2$ does not hold. Instead, we study an asymptotic sequence in which the number of approximating functions diverges, $k \to \infty$, and show that the bias terms vanish under explicit conditions. Concretely,[2]

$$\mathbb{E}[\hat{\beta}] - \beta = o(1), \quad \mathbb{E}[\hat{\sigma}_i^2] - \sigma_i^2 = o_p(1), \quad \mathbb{E}[\hat{\theta}] - \theta = o_p(1),$$

along sequences satisfying the assumptions below.

For these purposes, we assume that the unknown function $f(z)$ belongs to a class $\mathcal{F}$ of smooth functions that can be well approximated by linear combinations of the basis

2. For expositional purposes, we sometimes drop the Euclidean norm on the vector $\|v\|$, and write asymptotic results as, say, $v = o_p(1)$ instead of $\|v\| = o_p(1)$.

functions $p^1(z), \ldots, p^k(z)$.

Assumption 1. *We assume that $f \in \mathcal{F}$, where*

$$\mathcal{F} := \left\{ f : \min_{\alpha \in \mathbb{R}^k} \mathbb{E}\left[|f(z_i) - p_k(z_i)'\alpha|^2 \right] \leq C k^{-2\alpha_f}, \alpha_f > 1 \right\} \tag{5}$$

for some absolute constant $C < \infty$.

This implies that

$$\sup_{f \in \mathcal{F}} \mathbb{E}\left[|f(z_i) - p_k(z_i)'\hat{\alpha}_{-i}|^2 \right] = \mathcal{O}_p(k^{-2\alpha_f}),$$

so the mean-square approximation error decays at rate $k^{-2\alpha_f}$ uniformly over $\mathcal{F}$.

The constant α_f governs how fast approximation bias decays. To obtain consistency for the quadratic target studied here, we strengthen the usual semiparametric requirement from $\alpha_f > 0$ in Donald and Newey (1994) and Cattaneo, Jansson, and Newey (2018a) to $\alpha_f > 1$: compared with slope-type targets, quadratic forms require faster decay of approximation error. For polynomial or spline sieves on compact support with d_{cont} continuous covariates, $\alpha_f = s_f / d_{\text{cont}}$ where s_f is the number of continuous derivatives (Chen 2007), so the assumption reduces to $s_f > d_{\text{cont}}$: the unknown function must have more continuous derivatives than the number of continuous covariates entering flexibly. Discrete covariates such as gender, occupation, or region do not count toward d_{cont} because they are absorbed by group interactions rather than by the polynomial basis. In the wage applications of Section 8, the parsimonious sample has $d_{\text{cont}} = 1$ (age), so $s_f > 1$ suffices—the wage–age profile need only be twice continuously differentiable. The rich sample adds log revenue and log assets, giving $d_{\text{cont}} \leq 3$ and requiring $s_f > 3$. Both conditions are mild for smooth economic relationships. The condition is sufficient but not shown to be necessary.

Remark 1. *The results cover fixed-dimensional continuous covariates and fixed collections of discrete interaction groups, as in Section 8. They do not cover growing continuous dimension, growing numbers of interaction groups, or machine-learning nuisance classes.*

4. Consistency

In this section, we prove the consistency result for the proposed estimator $\hat{\theta}$. We study the asymptotic behavior of $\hat{\theta}$ assuming that x_i, z_i, and A are sequences of constants so that the only source of randomness is e_i. We adopt the conditional perspective of Scheffé (1959), Searle, Casella, and McCulloch (2009), and Kline, Saggio, and Sølvsten (2020), treating x_i, z_i, and A as fixed. This allows us to be agnostic about potential dependence between x_i, z_i, and A. Our analysis differs from Cattaneo, Jansson, and Newey (2018b), who consider sequences of random variables and condition on z_i only. Limits are taken assuming the number of observations goes to infinity, $n \to \infty$, the number of approximating functions

goes to infinity, $k \to \infty$ (so that the finite-sample bias of $\hat{\beta}$ and $\hat{\sigma}_i^2$ is asymptotically of negligible order), and the dimensionality of x_i goes to infinity, $p \to \infty$, to model the limited mobility bias. We make the following additional assumptions:

Assumption 2. *(i)* $\max_i(\mathbb{E}[e_i^4] + \sigma_i^{-2}) = \mathcal{O}(1)$*; (ii) there exists a* $c < 1$ *such that* $\max_i P_{W,ii} < c$ *for all* n*, where* $P_{W,ii} := I_n - M_{W,ii}$*; (iii)* $\max_i(x_i'\beta)^2 = \mathcal{O}(1)$.

Part *(i)* excludes heavy-tailed distributions of the errors as is typically assumed in the literature on OLS estimation (Cattaneo, Jansson, and Newey 2018b; Kline, Saggio, and Sølvsten 2020). Parts *(ii)* and *(iii)* coupled with the Assumption 1 imply that the leave-one-out estimator $\hat{\sigma}_i^2$ is well-defined and has bounded variance. Part *(ii)* also implies that $\frac{p+k}{n} \leq c < 1$ for all n.

Assumption 3. *We assume that* $k \to \infty$ *as* $n \to \infty$.

Shared rate restrictions for the proved results. The results below all rely on the same core asymptotics: the sieve dimension grows ($k \to \infty$), total regressor dimension remains bounded away from the sample size through $\max_i P_{W,ii} < c < 1$, and approximation bias decays fast enough through $\alpha_f > 1$. The theorem-specific conditions then add the restrictions needed for each object: Lemma 1 requires $\text{trace}(\tilde{A}^2) = o(1)$, Theorem 3 adds the fixed-rank leverage condition $\max_i v_i'v_i = o(1)$, and Theorem 4 adds the growing-rank negligibility conditions on $\max_i \left((\tilde{w}_i'\gamma)^2 + (\check{w}_i'\gamma)^2\right)$ and on the leading eigenvalue share $\lambda_1^2 / \sum_{\ell=1}^r \lambda_\ell^2$. In practical terms, k must grow fast enough for approximation error to vanish, but not so fast that the combined regressor dimension $p + k$ drives leverage close to one; this is the basic tradeoff behind the fixed-dimensional sieve results proved here.

Relative to linear leave-out estimation, replacing linear controls by a growing basis adds one additional task: one must show that the approximation terms from the unknown nuisance are negligible both for $\hat{\beta}$ and for the leave-out residual-variance correction. Lemma 1 and Theorems 3–4 establish exactly this under Assumptions 1–3.

Corollary 1. *Suppose the covariates entering the unknown function can be written as* $z_i = (u_i', s_i)'$*, where* $u_i \in \mathbb{R}^{d_{\text{cont}}}$ *has fixed dimension,* $s_i \in \{1, \ldots, G\}$ *indexes a fixed collection of discrete groups, and*

$$f(z_i) = \sum_{g=1}^{G} \mathbb{1}\{s_i = g\} f_g(u_i).$$

Assume the support of u_i *is compact and that, for the specified basis* $\{p^j(u_i)\}_{j=1}^k$*, each* f_g *satisfies*

$$\inf_{\alpha_g \in \mathbb{R}^k} \mathbb{E}\left[\left|f_g(u_i) - p_k(u_i)'\alpha_g\right|^2\right] \leq Ck^{-2s_f/d_{\text{cont}}}$$

for a common constant $C < \infty$ *and common smoothness index* $s_f > d_{\text{cont}}$*, uniformly over* g*. Then the interacted basis* $\{\mathbb{1}\{s_i = g\}p^j(u_i)\}_{g,j}$ *satisfies Assumption 1 with* $\alpha_f = s_f/d_{\text{cont}} > 1$*. If, in addition, Assumptions 2 and 3 hold together with the theorem-specific leverage and eigenvalue conditions, then the conclusions of Lemma 1, Theorem 3, and Theorem 4 apply to this interacted specification.*

Let $\tilde{A} := S_{xx}^{-1/2} A S_{xx}^{-1/2}$ and write its eigendecomposition as $\tilde{A} = Q\Lambda Q'$, where $\Lambda = \text{diag}(\lambda_1, \ldots, \lambda_r)$ collects the r nonzero eigenvalues and $Q \in \mathbb{R}^{p \times r}$ the corresponding eigenvectors.

Lemma 1. *If Assumptions 1, 2, and 3 hold, A is positive semi-definite, $\theta = \beta' A \beta = \mathcal{O}(1)$, and trace$(\tilde{A}^2) = \sum_{\ell=1}^{r} \lambda_\ell^2 = o(1)$, then*

$$\hat{\theta} - \theta \xrightarrow{p} 0.$$

5. Large-scale approximation

In this section we discuss an alternative estimator that allows for fast computation in typical large-scale applications such as those based on administrative linked employer-employee data. We follow Kline, Saggio, and Sølvsten (2020) by considering the random projection method of Achlioptas (2003). To describe it, fix $m \in \mathbb{N}$ and generate matrices $R_B, R_P \in \mathbb{R}^{m \times n}$ where each (i, j) coordinate is a random draw from the following Rademacher distribution:

$$R_{\cdot,ij} = \begin{cases} +1 & \text{with probability } 1/2, \\ -1 & \text{with probability } 1/2. \end{cases}$$

Decompose $A = 1/2(A_1' A_2 + A_2' A_1)$ for $A_1, A_2 \in \mathbb{R}^{n \times p}$, where $A_1 = A_2$ if A is positive semi-definite. Denote

$$\hat{P}_{W,ii} := \frac{1}{m} \left\| R_P W S_{ww}^{-1} w_i \right\|^2, \quad \hat{B}_{ii} := \frac{1}{m} \left(R_B A_1 S_{xx}^{-1} \sum_{j=1}^{n} M_{ij} x_j \right)' \left(R_B A_2 S_{xx}^{-1} \sum_{j=1}^{n} M_{ij} x_j \right).$$

The proposed estimator is then:

$$\hat{\theta}_{\text{JLA}} := \hat{\beta}' A \hat{\beta} - \sum_{i=1}^{n} \hat{B}_{ii} \hat{\sigma}_{i,\text{JLA}}^2, \quad \hat{\sigma}_{i,\text{JLA}}^2 := \frac{y_i (y_i - w_i' \hat{\gamma})}{1 - \hat{P}_{W,ii}} \left(1 - \frac{1}{m} \frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2}{1 - \hat{P}_{W,ii}} \right). \tag{6}$$

As in Kline, Saggio, and Sølvsten (2020), the term $\frac{1}{m} \frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2}{1 - \hat{P}_{W,ii}}$ removes a non-linearity bias from the approximation of $P_{W,ii}$ by $\hat{P}_{W,ii}$.

Lemma 2. *If Assumptions 1, 2, and 3 are satisfied, $n/m^4 = o(1)$, $\text{var}[\hat{\theta}]^{-1} = \mathcal{O}(n)$, and one of the following conditions hold, then $\text{var}[\hat{\theta}]^{-1/2}(\hat{\theta}_{JLA} - \hat{\theta} - \text{B}_m) = o_p(1)$ where $|\text{B}_m| \leq \frac{1}{m} \sum_{i=1}^{n} P_{W,ii}^2 |B_{ii}| \sigma_i^2$:*

(i) A is positive semi-definite and $\mathbb{E}[\hat{\beta}' A \hat{\beta}] - \theta = \sum_{i=1}^{n} B_{ii} \sigma_i^2 = \mathcal{O}(1)$.

(ii) $A = 1/2(A_1' A_2 + A_2' A_1)$ where $\theta_1 = \beta' A_1' A_1 \beta$ and $\theta_2 = \beta' A_2' A_2 \beta$ satisfy (i) and $\frac{\text{var}[\hat{\theta}_1] \, \text{var}[\hat{\theta}_2]}{n \, \text{var}[\hat{\theta}]^2} = \mathcal{O}(1)$.

6. Limiting distributions

This section derives two limiting distribution results for $\hat{\theta}$. The first assumes that the rank r of A is fixed; the second allows r to grow with n.

6.1. Fixed rank

First, we derive the limiting distribution of $\hat{\theta}$ for fixed r. Using the eigendecomposition $\tilde{A} = Q\Lambda Q'$ introduced before Lemma 1, we represent the estimator as

$$\hat{\theta} = \sum_{\ell=1}^{r} \lambda_\ell \left(\hat{b}_\ell^2 - \widehat{\text{var}}[\hat{b}_\ell] \right),$$

where $\hat{b} := \sum_{i=1}^n v_i y_i$, $\widehat{\text{var}}[\hat{b}] := \sum_{i=1}^n v_i v_i' \hat{\sigma}_i^2$, and $v_i := Q' S_{xx}^{-1/2} \sum_{j=1}^n M_{ij} x_j$.

The next theorem gives the fixed-rank limit for $\hat{\theta}$ and consistency of $\widehat{\text{var}}[\hat{b}]$.

Theorem 3. *If Assumptions 1, 2, and 3 hold, r is fixed, and $\max_i v_i' v_i = o(1)$, then*

(i) $\text{var}[\hat{b}]^{-1/2}(\hat{b} - b) \xrightarrow{d} \mathcal{N}(0, I_r)$, *where* $b := Q' S_{xx}^{1/2} \beta$,

(ii) $\text{var}[\hat{b}]^{-1} \widehat{\text{var}}[\hat{b}] \xrightarrow{p} I_r$,

(iii) $\hat{\theta} = \sum_{\ell=1}^r \lambda_\ell \left(\hat{b}_\ell^2 - \text{var}[\hat{b}_\ell] \right) + o_p(\text{var}[\hat{\theta}]^{1/2})$.

The growing-rank case is more relevant for AKM-type applications, so we turn to that setting next.

6.2. Growing rank

Define $\tilde{w}_i := \check{A} S_{ww}^{-1} w_i$, where $\check{A} := \begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix}$ and $\gamma := \begin{pmatrix} \beta & \alpha \end{pmatrix}'$. Then

$$\theta = \gamma' \check{A} \gamma = \gamma' S_{ww} S_{ww}^{-1} \check{A} \gamma = \sum_{i=1}^n \gamma' w_i \tilde{w}_i' \gamma$$

and the leave-one-out estimator can be written as

$$\hat{\theta} = \sum_{i=1}^n y_i \tilde{w}_i' \hat{\gamma}_{-i},$$

where the approximation error from replacing $\gamma' w_i$ by y_i is controlled by Assumption 1 and $k \to \infty$. A direct algebraic rearrangement yields

$$\hat{\theta} = \sum_{i=1}^n \sum_{\ell \neq i} C_{i\ell} y_i y_\ell,$$

where $C_{i\ell} := B_{W,i\ell} - 2^{-1} M_{W,i\ell} \left(B_{W,ii} M_{W,ii}^{-1} + B_{W,\ell\ell} M_{W,\ell\ell}^{-1} \right)$. Thus $\hat{\theta}$ is a second-order U-statistic with sample-dependent kernel $C_{i\ell}$. Theorem 4 combines this representation with

Appendix Lemma 3; relative to Kline, Saggio, and Sølvsten (2020), the only additional term is the sieve approximation error, which is negligible under Assumption 1 and $k \to \infty$.

Define also

$$\check{w}_i := \sum_{\ell=1}^{n} M_{W,i\ell} \frac{B_{W,\ell\ell}}{1 - P_{W,\ell\ell}} w_\ell.$$

Theorem 4. *If Assumption 1, 2, and 3 hold, and the following conditions are satisfied,*

$$\textit{(i)}\ \mathrm{var}[\hat{\theta}]^{-1} \max_i \left((\tilde{w}_i'\gamma)^2 + (\check{w}_i'\gamma)^2 \right) = o(1), \quad \textit{(ii)}\ \frac{\lambda_1^2}{\sum_{\ell=1}^{r} \lambda_\ell^2} = o(1),$$

then $\mathrm{var}[\hat{\theta}]^{-1/2}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, 1)$.

7. Simulation study

We use a Monte Carlo study to evaluate the estimator under controlled conditions that isolate the mechanisms emphasized in the theory. In the baseline design, the nuisance covariates z_i are drawn uniformly on a fixed cube, the linear regressors x_i are nonlinear functions of z_i, and the target quadratic form is $\theta = \beta' A \beta$, where A selects the first two coordinates of β, so the target measures dispersion in two fixed effects. Unless otherwise noted, the nonlinear designs use $n = 500$, nuisance dimension $d = 4$, and $p = 90$ linear regressors. Across designs, we vary the nonlinearity of $f(z)$, the error distribution, the dimension of the nuisance, the number of linear regressors, leverage, and sample size.

We compare the plug-in estimator with leave-out estimators that use degree-1, degree-3, and degree-5 series approximations. Cubic polynomials are common in applied work (Card et al. 2018); the degree-5 basis lets us check whether going beyond cubic matters. Table 1 reports the subset of designs that span the main mechanisms in the paper: approximation error and nonlinearity, robustness to heteroskedasticity and heavy tails, and finite-sample stress from nuisance dimension, many regressors, leverage, and sample size. The first four columns report absolute bias for the plug-in (PI) and leave-out (LOO) estimators at each basis degree. The last two report RMSE and nominal 95% delta-method coverage for LOO at degree 5, so the table separates point-estimation performance from finite-sample inference.

Approximation error and the smoothness condition. Assumption 1 predicts that sieve approximation error decays at rate $k^{-2\alpha_f}$, so richer bases should reduce bias most when f departs strongly from linearity. The simulations confirm this. In the strong-nonlinearity design, absolute bias falls from 0.074 for PI(1) to 0.046–0.047 for the degree-3 and degree-5 leave-out estimators; under high leverage the reduction is even larger, from 0.086 to 0.041. By contrast, in the linear benchmark and the mildly nonlinear design, all estimators perform similarly—as expected when approximation error is already small. The first panel of Table 1 should therefore be read as evidence on the practical bite of the smoothness

Table 1: Simulation study: results

Scenario	PI(1)	LOO(1)	LOO(3)	LOO(5)	RMSE(5)	Cov.(5)
Approximation and nonlinearity						
Mild nonlinearity	0.047	0.048	0.049	0.051	0.063	0.90
Linear benchmark	0.029	0.028	0.030	0.033	0.040	0.77
Strong nonlinearity	0.074	0.068	0.046	0.047	0.059	0.90
Robustness stress tests						
Heteroskedasticity	0.077	0.081	0.074	0.073	0.092	0.79
Heavy-tailed errors	0.055	0.057	0.056	0.056	0.070	0.86
Higher nuisance dimension	0.049	0.045	0.038	0.042	0.054	0.79
Many regressors	0.020	0.027	0.027	0.028	0.034	0.68
High leverage	0.086	0.076	0.046	0.041	0.052	0.69
Larger sample	0.022	0.022	0.020	0.020	0.026	0.81

Note: Columns PI(1), LOO(1), LOO(3), and LOO(5) report absolute bias for the plug-in estimator with a degree-1 basis and the leave-out estimator with degree-1, degree-3, and degree-5 bases, respectively. RMSE(5) and Cov.(5) report the RMSE and nominal 95% delta-method coverage of the degree-5 leave-out estimator. The displayed scenarios are chosen to cover the paper's main mechanisms: approximation error, heteroskedasticity and tail behavior, nuisance dimension, many-regressor stress, leverage, and sample size. Entries are based on 200 Monte Carlo replications for each design.

condition: basis enrichment pays off when misspecification is genuine, and costs little when it is not.

Heteroskedasticity and error tails. The leave-out variance correction in (2) relies on observation-specific residual variances $\hat{\sigma}_i^2$, whose quality depends on the moment conditions in Assumption 2*(i)*. In the heteroskedastic design, the degree-1 leave-out estimator is slightly more biased than PI(1) (0.081 versus 0.077): estimating heterogeneous σ_i^2 adds noise that can offset the correction's gains. Richer bases reduce bias only modestly, to 0.073 at degree 5. Heavy-tailed errors show a similar pattern, with bias around 0.055–0.057 regardless of basis degree. These designs demonstrate that heteroskedasticity and tail behavior create residual bias on a margin orthogonal to approximation error: basis enrichment alone does not address it.

Leverage and many-regressor stress. Assumption 2*(ii)* requires $\max_i P_{W,ii} < c < 1$, bounding individual leverage away from one as the combined regressor dimension $p + k$ grows. The many-regressor and high-leverage designs stress this condition. Point estimation remains reliable: in the many-regressor design, the degree-5 leave-out estimator has absolute bias of 0.028 and RMSE of 0.034. Finite-sample inference, however, deteriorates. Nominal 95% delta-method coverage is only 0.68 under many regressors, 0.69 under high leverage, 0.79 under higher nuisance dimension, and 0.79 under heteroskedasticity. The asymptotic approximation is slowest precisely when leverage and residual-variance estimation are most unstable—the conditions that the theory identifies as governing the rate restrictions in Lemma 1 and Theorems 3–4. Larger samples help: absolute bias falls to 0.020 and RMSE to 0.026, though coverage still reaches only 0.81.

Summary. For applied work, basis enrichment is most valuable when nonlinearity interacts with leverage or many regressors—the joint stress emphasized by the theory. When those stresses are absent, additional flexibility costs little but gains little. The estimator is most reliable as a point-estimation and decomposition tool; interval estimates should be interpreted with caution in demanding designs unless sample size is very large and leverage is well behaved.

8. Empirical application

We apply our estimator to decompose wage dispersion in linked employer–employee data. The exercise illustrates that relaxing linearity in the covariates yields stable and interpretable variance components, in contrast to the standard linear AKM model once rich controls are introduced.

8.1. Data and estimation

Our empirical analysis combines two Portuguese sources of administrative data. The matched employer–employee dataset *Quadros de Pessoal* (QP) reports worker characteristics (age, gender, occupation, qualification, education, contract type, hours, compensation) and firm characteristics (location, industry, revenues) for the universe of private-sector firms. The firm accounts dataset *Central de Balanços* (CB) provides annual balance-sheet and income-statement information (including total assets and sales) for non-financial corporations. We link QP and CB at the firm level to obtain a panel with rich worker- and firm-side observables.

We focus on the period 2008–2018 and on full-time workers aged 20–65. The model's dependent variable is the logarithm of individual hourly wages. As in the typical "AKM" literature, we decompose the cross-sectional variance of wages into the variance of worker fixed effects (σ_α^2), the variance of firm fixed effects (σ_ψ^2), and their covariance $\text{cov}(\alpha, \psi)$, which captures sorting.

We estimate a nested five-step specification ladder within each panel, progressively relaxing the restrictions on how covariates enter the wage equation: (i) a pure AKM baseline with no observed controls beyond the fixed effects; (ii) additive linear controls; (iii) heterogeneous linear controls, which allow slopes on continuous variables to differ across demographic groups; (iv) nonlinear homogeneous controls, which replace the linear terms with a common degree-5 polynomial basis; and (v) the full model, which interacts the polynomial basis with demographic groups. The ladder separates the roles of adding observables, allowing slope heterogeneity, and allowing nonlinearity. The polynomial bases are interacted with a fully saturated set of observed worker categorical variables (gender in Panel A; gender $\times$ education $\times$ qualification in Panel B), so the full model allows group-specific nonlinear effects of each continuous covariate.

To connect this specification to the theoretical framework of Section 2, let $s_i \in$

$\{1, \ldots, G\}$ denote the worker categorical group. The group-specific model

$$y_i = x_i'\beta + \sum_{s=1}^{G} \mathbb{1}\{s_i = s\} f_s(z_i^{\text{cont}}) + e_i$$

is a special case of (1) with the unknown function redefined on the expanded argument $\tilde{z}_i := (z_i^{\text{cont}\prime}, s_i)'$. The natural basis for this piecewise-smooth function is the set of indicator-weighted polynomials $\{\mathbb{1}\{s_i = s\} \cdot p^j(z_i^{\text{cont}})\}_{s,j}$, which is exactly the interacted basis P_k used in our implementation. Because the discrete indicators partition the sample rather than adding continuous dimensions, the smoothness condition (Assumption 1) applies within each group at rate $\alpha_f = s_f / d_{\text{cont}}$, where d_{cont} is the number of continuous covariates and s_f is the number of their continuous derivatives. This is exactly the fixed-dimensional interacted-sieve case formalized in Corollary 1. In Panel A, $d_{\text{cont}} = 1$ because age is the only continuous covariate entering f; in Panel B, $d_{\text{cont}} = 3$ because the generalized nuisance depends on age, revenue, and assets, while the interaction groups remain fixed collections of observed worker categories. Note that the interaction groups consist of worker categorical variables, not firm identifiers: including firm identifiers would make P_k nearly collinear with the firm fixed-effect indicators in X, jeopardizing the full-rank condition on $S_{xx} = X'MX$ required by Assumption 2.

We report estimates under a parsimonious control set (gender, age) and an extended set that adds education and qualification at the worker level and revenue and assets at the firm level.[3]

Detailed sample descriptives, connected-component coverage, and empirical ladder diagnostics are reported in Appendix Tables 6, 7, and 8.

Table 2 characterizes the firm-level controls that distinguish the two panels. Within-firm time variation in logs is limited: about 18.8% for revenue, 6.4% for assets, and 7.1% for workers. Missingness is material though declining for revenue and assets (roughly 26% in 2008 to 18% in 2018), while workers missingness is 0% as we impute back missing employment from the matched employer-employee data. Because most of the variation in these controls is cross-sectional, adding them as linear regressors in the presence of firm fixed effects introduces near-collinearity—a point that will be central to interpreting the results below.

8.2. Results

Table 3 reports the bias-corrected variance components at each rung of the ladder. Table 4 reports within-panel differences between successive rungs; the transitions branch from the linear additive specification, so one path adds heterogeneity first and the other adds nonlinearity first, and both paths arrive at the full model. Table 5 reports the variance of the residualized wage (after partialling out observables) as a share of total wage variance:

3. In practice, we first partial out the covariates, then compute the bias-corrected variance components on the residualized wage. Age is normalized at 50 throughout, and the common linear age term is excluded because it is not identified in the presence of worker and year fixed effects. All variance components in Tables 3 and 4 are reported as shares of total log-wage variance.

Table 2: Firm controls: within-firm time variation and missingness

	Non-missing firm-year	Firms (≥2y)	Mean years (firms ≥2y)	Within share (log) (%)	Zero within SD (%)
A. Firm-control time variation in estimation sample					
Revenue	1,874,310	290,932	6.22	18.79%	1.67%
Assets	1,874,310	290,932	6.22	6.43%	0.72%
Workers	2,397,757	383,679	5.99	7.13%	27.11%
B. Missingness by year (firm-year panel)					
Year	Firm-years	Revenue miss.	Assets miss.	Workers miss.	
2008	253,005	26.27%	26.27%	0.00%	–
2009	245,077	25.37%	25.37%	0.00%	–
2010	223,267	23.66%	23.66%	0.00%	–
2011	218,229	23.01%	23.01%	0.00%	–
2012	202,744	22.47%	22.47%	0.00%	–
2013	198,597	21.20%	21.20%	0.00%	–
2014	202,625	20.36%	20.36%	0.00%	–
2015	207,037	19.89%	19.89%	0.00%	–
2016	211,892	19.39%	19.39%	0.00%	–
2017	215,908	18.80%	18.80%	0.00%	–
2018	219,376	18.24%	18.24%	0.00%	–

Note: Panel A reports non-missing firm-year observations and within-firm time variation of firm controls in the estimation sample (2008–2018), restricting the within-variation calculations to firms observed for at least two years. The within-share statistic is the share of total log variance attributable to within-firm over-time variation. Panel B reports year-by-year missingness rates for the same firm controls in the firm-year panel; worker counts use a MEE-based fallback when SCIE firm controls are missing.

values above one mean that the linear partialling-out step amplifies rather than removes variance, a direct symptom of collinearity between controls and fixed effects. Because Panel B is estimated on the smaller sample with non-missing firm controls, the informative comparisons are the within-panel moves along the ladder, not the level differences across panels.

The basic pattern is clearest within Panel A. The AKM baseline, heterogeneous linear, nonlinear homogeneous, and full specifications all lie in a relatively narrow range: worker variance stays around 0.50, firm variance around 0.15, and sorting around 0.09. The additive linear specification is the outlier, with worker variance rising to 0.7175 and firm variance to 0.1818. This is the only specification in Panel A that forces men and women to share the same age-wage profile and restricts its shape to a low-order polynomial without higher-degree terms. Table 4 shows that either allowing gender-specific slopes or allowing a richer common polynomial largely reverses the increase. In the parsimonious panel, then, flexibility matters at the margin, but the gains are moderate and the different flexible specifications tell a very similar story.

The central contrast appears in Panel B. Once education, qualification, revenue, and assets are added linearly, the decomposition becomes unstable: σ_α^2 jumps to 2.8076, σ_ψ^2 to 0.9419, and sorting turns negative at -0.1389. Table 5 shows the same point from the accounting side: the variance of the residualized wage rises to 4.1251 times total wage variance under the linear additive specification. A ratio above one means that the linear

Table 3: Wage variance decomposition

Model	σ^2_α	σ^2_ψ	$\mathrm{cov}(\alpha, \psi)$
Panel A: Parsimonious controls			
AKM baseline	0.5173	0.1596	0.0989
Linear additive controls	0.7175	0.1818	0.0815
Heterogeneous linear controls	0.4976	0.1489	0.0907
Nonlinear homogeneous controls	0.5205	0.1474	0.0952
Full model	0.5007	0.1482	0.0895
Panel B: Extended controls			
AKM baseline	0.5400	0.1575	0.0851
Linear additive controls	2.8076	0.9419	−0.1389
Heterogeneous linear controls	0.5619	1.1174	−0.0617
Nonlinear homogeneous controls	0.2999	0.1420	0.0187
Full model	0.2344	0.1301	0.0107

Note: Entries report bias-corrected (KSS) variance components as shares of total log-wage variance: worker-effect variance (σ^2_α / var(y)), firm-effect variance (σ^2_ψ / var(y)), and their covariance (cov(α, ψ) / var(y)). "Linear additive controls" use additive observed controls, with age entering through the standard normalized profile centered at 50. "Heterogeneous linear controls" allow group-specific linear deviations. "Nonlinear homogeneous controls" add a common degree-5 polynomial basis, and "Full model" interacts that basis with worker groups. Values above one are possible because bias-corrected components need not sum to one.

partialling-out step is introducing more noise than it removes, which is what happens when nearly collinear regressors produce unstable coefficient estimates. Allowing heterogeneous linear slopes does not solve the problem: worker variance falls back to 0.5619, but firm variance remains implausibly high at 1.1174, sorting stays negative, and the residualized-wage variance ratio is still 1.6686.

The correction comes from nonlinearity. Moving from linear additive to nonlinear homogeneous controls lowers worker variance by 2.5077, firm variance by 0.8000, and raises sorting by 0.1576, bringing the decomposition back to plausible magnitudes; the residualized-wage variance ratio falls from 4.1251 to 0.6458. Once that nonlinear correction is in place, adding group-specific interactions has a second-order effect: worker variance falls from 0.2999 to 0.2344, firm variance from 0.1420 to 0.1301, and the residualized-wage variance ratio from 0.6458 to 0.5563.

Why does the linear specification break down so sharply in Panel B? Revenue and assets are predominantly cross-sectional: as documented in Table 2, within-firm variance shares are only 6–19%. Adding them as linear regressors alongside firm fixed effects therefore creates severe collinearity. In the bias-corrected estimator, that inflates the leverage and influence terms entering the correction, amplifying rather than removing instability. The ladder shows that the main cure is not simply heterogeneous slopes; it is allowing the data to depart from a restrictive linear functional form for those controls.

Overall, the relevant empirical contrast is not "linear" versus "flexible" in the abstract, but which margin of flexibility matters. In these data, nonlinearity does most of the work, especially once rich firm controls are included. Heterogeneity refines the decomposition further, but it does not by itself eliminate the instability created by the linear additive

Table 4: Stepwise changes in wage variance decomposition

Change	$\Delta\sigma^2_\alpha$	$\Delta\sigma^2_\psi$	$\Delta\,\mathrm{cov}(\alpha,\psi)$
Panel A: Parsimonious controls			
Add linear observables	0.2002	0.0222	−0.0174
Add heterogeneity at degree 1	−0.2200	−0.0329	0.0092
Add nonlinearity without interactions	−0.1970	−0.0344	0.0138
Add heterogeneity at degree 5	−0.0198	0.0008	−0.0057
Add nonlinearity with interactions	0.0032	−0.0006	−0.0012
Panel B: Extended controls			
Add linear observables	2.2676	0.7845	−0.2239
Add heterogeneity at degree 1	−2.2457	0.1755	0.0772
Add nonlinearity without interactions	−2.5077	−0.8000	0.1576
Add heterogeneity at degree 5	−0.0655	−0.0119	−0.0080
Add nonlinearity with interactions	−0.3274	−0.9874	0.0723

Note: Entries report within-panel differences in the bias-corrected variance components when moving from one rung of the specification ladder to the next. These are descriptive model-comparison deltas, not formal tests.

specification.

8.3. Interpreting the role of flexible controls

The within-panel ladder suggests that part of what appears as worker–firm sorting or inflated firm premia in linear AKM reflects misspecified returns to observed characteristics rather than a genuine correlation between unobserved worker and firm types. The full specification allows the return to age, revenue, or assets to differ across demographic groups (gender in Panel A; gender × education × qualification in Panel B), so the effective covariate adjustment varies — for instance, the return to working at a high-revenue firm can differ between high-educated men and low-educated women. Once this flexibility is allowed, both fixed-effect variances and the covariance term fall, especially in the panel with firm controls.

Several caveats apply. First, this conclusion speaks only to heterogeneity along observed z_{it}. Match-specific premia or complementarities outside z_{it} can still matter (e.g., a manager–worker fit that raises pay on a particular team). Second, because the interaction groups are defined by worker categories rather than firm identifiers, the model does not estimate firm-specific pay schedules *per se*. Firms differ in their effective covariate adjustment only to the extent that they employ different mixes of worker types and have different levels of the continuous firm controls. Allowing firm-level interaction groups is a natural extension that we leave for future work. Third, our decomposition is static: time-varying firm policies or match dynamics can exist without affecting the estimated variance components.

Table 5: Variance remaining after partialling out observables

Model	$\mathrm{var}(\tilde{y}) / \mathrm{var}(y)$
Panel A: Parsimonious controls	
AKM baseline	0.9976
Linear additive controls	1.3022
Heterogeneous linear controls	0.9516
Nonlinear homogeneous controls	0.9792
Full model	0.9552
Panel B: Extended controls	
AKM baseline	0.9973
Linear additive controls	4.1251
Heterogeneous linear controls	1.6686
Nonlinear homogeneous controls	0.6458
Full model	0.5563

Note: Entries report $\mathrm{var}(\tilde{y}) / \mathrm{var}(y)$, where $\tilde{y}$ is the wage after partialling out observables and year effects, before the KSS decomposition is applied. This is a descriptive accounting object, not an orthogonal variance decomposition. Values above one indicate that the linear partialling-out step is itself unstable.

9. Conclusions

Standard AKM wage decompositions assume that observed covariates enter log wages linearly. This paper has shown, both theoretically and empirically, that this restriction can distort the decomposition when rich controls are included.

On the theory side, we developed a semiparametric leave-out estimator that replaces the linear control function with an unknown smooth function approximated by a growing basis. The estimator preserves the original variance-component targets of AKM while allowing the data to determine how covariates affect wages. We established consistency and asymptotic normality under heteroskedasticity and many fixed effects, identifying the smoothness condition that separates quadratic targets from the linear functionals studied in the existing semiparametric series literature. The required smoothness is stronger—the unknown function must have more continuous derivatives than continuous covariates—but is mild in typical wage applications.

On the empirical side, the application to Portuguese administrative data showed that the standard linear AKM decomposition becomes unstable when firm-level controls such as revenue and assets are added: worker-effect and firm-effect variances inflate by an order of magnitude and the sorting term reverses sign. Estimating five increasingly flexible specifications revealed that the main correction comes from allowing nonlinearity in how these controls enter the wage equation, not from allowing heterogeneous slopes across demographic groups. In the parsimonious specification with only gender and age, the decomposition was stable throughout, confirming that the instability is specific to the interaction of predominantly cross-sectional firm controls with a linear functional form. The broader lesson is that apparent sorting or inflated firm premia in linear AKM can

partly reflect misspecified returns to observed characteristics.

Several directions remain open. Our theory covers fixed-dimensional sieve designs; extending it to growing continuous dimension, approximate sparsity, or machine-learning nuisance estimators would broaden the class of specifications the estimator can accommodate. The interaction groups in our empirical specification are defined by worker categories; allowing firm-level interactions would permit firm-specific pay schedules, though at the cost of near-collinearity with firm fixed effects. Finally, the simulations showed that finite-sample inference can be imperfect even when point estimation is reliable, particularly under high leverage. Improving the coverage properties of confidence intervals in empirically demanding designs is a natural next step.

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A. Proofs

Here, we provide proofs of the main results found in the paper and additional derivations.

Proof of Lemma 1. The variance of $\hat{\beta}$ is

$$\begin{aligned}
\operatorname{var}[\hat{\beta}] &= \operatorname{var}\left[S_{xx}^{-1}\sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij}x_i f(z_j) + S_{xx}^{-1}\sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij}x_i e_j\right] \\
&= S_{xx}^{-1}\operatorname{var}\left[\sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij}x_i f(z_j) + \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij}x_i e_j\right] S_{xx}^{-1} \\
&= S_{xx}^{-1}\left(\sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij}^2 x_i x_i' \sigma_j^2\right) S_{xx}^{-1}
\end{aligned}$$

under the assumption of fixed x_i and z_i.

First, rewrite the difference between the estimator and the estimand as

$$\begin{aligned}
\hat{\theta} - \theta &= \hat{\beta}' A \hat{\beta} - \beta' A \beta - \sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2 \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} y_j x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} x_i y_j - \beta' A \beta - \sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2.
\end{aligned}$$

Given a data-generating process for y_i, we can expand $\hat{\theta} - \theta$ further as

$$\begin{aligned}
\hat{\theta} - \theta &= \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij}\left(x_j'\beta + f(z_j) + e_j\right) x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} x_i \left(x_j'\beta + f(z_j) + e_j\right) - \beta' A \beta - \sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2 \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} x_j' \beta x_i S_{xx}^{-1} A S_{xx}^{-1} \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} x_i x_j' \beta + \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} f(z_j) x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} x_i f(z_j) \\
&+ \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} e_j x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} x_i e_j + \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} x_j' \beta x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} x_i f(z_j) \\
&+ \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} f(z_j) x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} x_i x_j' \beta + \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} x_j' \beta x_i S_{xx}^{-1} A S_{xx}^{-1} \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} x_i e_j \\
&+ \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} e_j x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} x_i x_j' \beta + \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} f(z_j) x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} x_i e_j \\
&+ \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} e_j x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} x_i f(z_j) - \beta' A \beta - \sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2.
\end{aligned}$$

Next, we use the definition $B_{i\ell} := \sum_{j=1}^{n} M_{ij} x_j' S_{xx}^{-1} A S_{xx}^{-1} \sum_{j=1}^{n} M_{\ell j} x_j$, so that

$$\begin{aligned}
\hat{\theta} - \theta &= \sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} f(z_i)^2 + \sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} e_i^2 - \sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2 \\
&+ 2\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} x_\ell' \beta f(z_i) + 2\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} x_\ell' \beta e_i + 2\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} e_\ell f(z_i).
\end{aligned}$$

Finally, rearranging, we have

$$\begin{aligned}\hat{\theta} - \theta = &\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} f(z_i)^2 + \sum_{i=1}^{n}\sum_{\ell\neq i} B_{i\ell} e_i e_\ell + \sum_{i=1}^{n} B_{ii}(e_i^2 - \hat{\sigma}_i^2) \\ &+ 2\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} x_\ell' \beta f(z_i) + 2\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} x_\ell' \beta e_i + 2\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} e_\ell f(z_i).\end{aligned} \tag{7}$$

In general, it holds that

$$\mathbb{E}[|\hat{\theta} - \theta|^2] = |\mathbb{E}[\hat{\theta} - \theta]|^2 + \text{trace}(\text{var}[\hat{\theta} - \theta]).$$

To show that $\hat{\theta}$ is consistent for θ, we need to show that the bias and the variance of the difference in (7) goes to zero. The main idea is to compute bounds on each term and their variances, and show that these bounds are asymptotically negligible. Then, convergence in the quadratic mean would imply convergence in probability.

Applying expectations on both sides, and using independence and mean-zero properties of errors, we have

$$\mathbb{E}[\hat{\theta} - \theta] = \sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} f(z_i)^2 + 2\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} x_\ell' \beta f(z_i) + \sum_{i=1}^{n} B_{ii}\mathbb{E}[\sigma_i^2 - \hat{\sigma}_i^2].$$

We now prove that each term goes to zero in probability as $n \to \infty$, $k \to \infty$, and $p \to \infty$.

Denote $B := (B_{i\ell})_{i,\ell=1}^{n} \in \mathbb{R}^{n\times n}$, and $F := (f(z_1), \ldots, f(z_n))' \in \mathbb{R}^n$. Then

$$\begin{aligned}\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} f(z_i)^2 &= F'BF \\ &= F'MXS_{xx}^{-1}AS_{xx}^{-1}X'MF \\ &= F'MXS_{xx}^{-1}A^{1/2}A^{1/2}S_{xx}^{-1}X'MF \\ &= (F'MXS_{xx}^{-1}A^{1/2})^2,\end{aligned}$$

where we use $A = A^{1/2}A^{1/2}$ because A is symmetric. By the Markov inequality, assumptions on $\mathcal{F}$, M being idempotent, and the Cauchy-Schwartz inequality,

$$\begin{aligned}\left\|\frac{1}{n}F'MXS_{xx}^{-1}A^{1/2}\right\| &\leq \text{trace}\left(\frac{1}{n}F'MF\right)^{1/2} \cdot \text{trace}\left(\frac{1}{n}A^{1/2}S_{xx}^{-1}X'MXS_{xx}^{-1}A^{1/2}\right)^{1/2} \\ &= \text{trace}\left(\frac{1}{n}F'MF\right)^{1/2} \cdot \text{trace}\left(\frac{1}{n}A^{1/2}S_{xx}^{-1}A^{1/2}\right)^{1/2} \\ &= \mathcal{O}(k^{-\alpha_f}\sqrt{p/n}).\end{aligned}$$

Thus, using the Assumption 1, we have

$$\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} f(z_i)^2 = \mathcal{O}(k^{-2\alpha_f} np) = \mathcal{O}(k^{-\alpha_f} n) \to 0.$$

Now, for the second term in the main decomposition,

$$\begin{aligned}\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} x_\ell' \beta f(z_i) &= F'BX\beta \\ &= F'MXS_{xx}^{-1}AS_{xx}^{-1}X'MX\beta \\ &= F'MXS_{xx}^{-1}A\beta.\end{aligned}$$

Given that $\beta' AS_{xx}^{-1}A\beta = \mathcal{O}(1)$, by the Markov inequality, assumption on $\mathcal{F}$, M being idempotent, and the Cauchy-Schwartz inequality we have

$$\begin{aligned}\left\|\frac{1}{n}F'MXS_{xx}^{-1}A\beta\right\| &\leq \text{trace}\left(\frac{1}{n}F'MF\right)^{1/2} \cdot \text{trace}\left(\frac{1}{n}\beta' AS_{xx}^{-1}X'MXS_{xx}^{-1}A\beta\right)^{1/2} \\ &= \text{trace}\left(\frac{1}{n}F'MF\right)^{1/2} \cdot \text{trace}\left(\frac{1}{n}\beta' AS_{xx}^{-1}A\beta\right)^{1/2} \\ &= \mathcal{O}(k^{-\alpha_f}/\sqrt{n}).\end{aligned}$$

So that using the Assumption 1, we have

$$\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} x_\ell' \beta f(z_i) = \mathcal{O}(k^{-\alpha_f}\sqrt{n}) \to 0.$$

To bound the third term, we should show that

$$\mathbb{E}[\hat{\sigma}_i^2 - \sigma_i^2] = \left(x_i'\beta + f(z_i)\right)\left(x_i'\mathbb{E}[\beta - \hat{\beta}_{-i}] + \mathbb{E}[f(z_i) - p_k(z_i)'\hat{\alpha}_{-i}]\right)$$

goes in probability to zero. This is equivalent to bounding the bias of the estimator of the slope coefficient and the bias of the function approximation.

The bias of the slope estimator is

$$\mathbb{E}[\hat{\beta} - \beta] = \mathbb{E}\left[\left(\frac{1}{n}X'MX\right)^{-1}\frac{1}{n}X'MF\right].$$

By the Markov inequality, assumption on $\mathcal{F}$, M being idempotent, and the Cauchy-Schwartz inequality we have that:

$$\begin{aligned}\left\|\frac{1}{n}X'MF\right\| &\leq \text{trace}\left(\frac{1}{n}X'MX\right)^{1/2} \cdot \text{trace}\left(\frac{1}{n}F'MF\right)^{1/2} \\ &= \mathcal{O}(k^{-\alpha_f}\sqrt{p/n}),\end{aligned}$$

and, similarly,

$$\left\|\frac{1}{n}X'MX\right\| = \mathcal{O}(p/n),$$

so that using Assumption 1, we ultimately have:

$$\mathbb{E}[\|\hat{\beta} - \beta\|] = \mathcal{O}(\sqrt{n}/k^{\alpha_f}\sqrt{p}) \to 0.$$

Because of the assumption on the functional class $\mathcal{F}$, we can bound the bias of the function approximation using Jensen's inequality for convex $x \mapsto x^2$,

$$\begin{aligned}\mathbb{E}[|f(z_i) - p_k(z_i)'\hat{\alpha}_{-i}|] &\leq \mathbb{E}[|f(z_i) - p_k(z_i)'\hat{\alpha}_{-i}|^2]^{1/2} \\ &\leq (Ck^{-2\alpha_f})^{1/2} = \mathcal{O}_p(k^{-\alpha_f}),\end{aligned}$$

so that:

$$\mathbb{E}[|\hat{\sigma}_i^2 - \sigma_i^2|] = \mathcal{O}_p(\sqrt{n}/k^{\alpha_f}\sqrt{p}) \xrightarrow{p} 0.$$

From this it follows that:

$$\mathbb{E}[|\hat{\theta} - \theta|] \xrightarrow{p} 0.$$

We next turn our attention to bounding the variance. Let matrix

$$\tilde{A} := S_{xx}^{-1/2} A S_{xx}^{-1/2},$$

and let $\lambda_1, \ldots, \lambda_r$ be its nonzero eigenvalues. We assume that $\lambda_1^2 \geq \ldots \geq \lambda_r^2$, and that each eigenvalue appears as many times as its algebraic multiplicity. Under these assumptions, we can spectrally decompose $\tilde{A} = QDQ'$, where Q is a matrix of orthonormal vectors and $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_r)$.

The variance of $2\sum_{i=1}^n \sum_{\ell=1}^n B_{i\ell} x_\ell' \beta e_i$ is

$$\begin{aligned}4\sum_{i=1}^n \left(\sum_{\ell=1}^n B_{i\ell} x_\ell' \beta\right)^2 \sigma_i^2 &\leq \max_i \sigma_i^2 \beta' X' B^2 X \beta = \max_i \sigma_i^2 \beta' A S_{xx}^{-1} A \beta \\ &\leq \max_i \sigma_i^2 \lambda_1 \theta = o(1).\end{aligned}$$

To explain why the last inequality holds, define $\tilde{\beta} = S_{xx}^{1/2}\beta$ so that $\beta = S_{xx}^{-1/2}\tilde{\beta}$,

$$\beta' A S_{xx}^{-1} A \beta = \tilde{\beta}' S_{xx}^{-1/2} A S_{xx}^{-1/2} \tilde{\beta} = \tilde{\beta}' (S_{xx}^{-1/2} A S_{xx}^{-1/2})^2 \tilde{\beta} = \tilde{\beta}' \tilde{A}^2 \tilde{\beta},$$

and

$$\theta = \tilde{\beta}' S_{xx}^{-1/2} A S_{xx}^{-1/2} \tilde{\beta} = \tilde{\beta}' \tilde{A} \tilde{\beta}.$$

Then, using the Rayleigh quotient argument, it holds that

$$\frac{\tilde{\beta}' \tilde{A}^2 \tilde{\beta}}{\tilde{\beta}' \tilde{A} \tilde{\beta}} \leq \lambda_{\max}(\tilde{A}),$$

because $\tilde{A}$ is positive semi-definite, $\tilde{\beta} \neq 0$, and $\lambda_{\max}(\tilde{A})$ is the largest eigenvalue of $\tilde{A}$. From this it follows that we can bound

$$\beta' A S_{xx}^{-1} A \beta \leq \lambda_1 \theta,$$

use assumptions $\theta = \mathcal{O}(1)$, $\lambda_1 \leq \mathrm{trace}(\tilde{A}^2)^{1/2} = o(1)$, and the fact that the variance is bounded. The conclusion then follows.

The variance of $2\sum_{i=1}^n \sum_{\ell=1}^n B_{i\ell} e_\ell f(z_i)$ is

$$4\sum_{i=1}^n \sum_{\ell=1}^n B_{i\ell}^2 \sigma_\ell^2 f(z_i)^2 \leq \max_i 4\sigma_i^2 f(z_i)^2 \sum_{i=1}^n \sum_{\ell=1}^n B_{i\ell}^2 = \max_i 4\sigma_i^2 f(z_i)^2 \text{trace}(\tilde{A}^2) = o(1),$$

which follows since

$$\begin{aligned}
\sum_{i=1}^n \sum_{\ell=1}^n B_{i\ell}^2 &= \|B\|_F^2 = \text{trace}(B'B) \\
&= \text{trace}(MXS_{xx}^{-1}AS_{xx}^{-1}AS_{xx}^{-1}X'M) \\
&= \text{trace}(MXS_{xx}^{-1/2}S_{xx}^{-1/2}AS_{xx}^{-1/2}S_{xx}^{-1/2}AS_{xx}^{-1/2}S_{xx}^{-1/2}X'M) \\
&= \text{trace}(S_{xx}^{-1/2}AS_{xx}^{-1/2}S_{xx}^{-1/2}AS_{xx}^{-1/2}S_{xx}^{-1/2}X'MXS_{xx}^{-1/2}) \\
&= \text{trace}(S_{xx}^{-1/2}AS_{xx}^{-1/2}S_{xx}^{-1/2}AS_{xx}^{-1/2}) = \text{trace}(\tilde{A}^2),
\end{aligned}$$

and we use the assumptions as above but now, instead of the bounded variance, we assume that $\mathcal{F}$ is a class of bounded functions so that $\max|f(z_i)| < C$ for $i = 1, \ldots, n$ for some absolute constant C (this is implied by the main assumption on $\mathcal{F}$).

Because $B_{i\ell}^2 = B_{\ell i}^2$ for any i, ℓ, the variance of $\sum_{i=1}^n \sum_{\ell \neq i} B_{i\ell} e_i e_\ell$ is

$$2\sum_{i=1}^n \sum_{\ell \neq i} B_{i\ell}^2 \sigma_i^2 \sigma_\ell^2 \leq \max_i 2\sigma_i^4 \sum_{i=1}^n \sum_{\ell=1}^n B_{i\ell}^2 = \max_i 2\sigma_i^4 \text{trace}(\tilde{A}^2) = o(1).$$

To compute the variance of $\sum_{i=1}^n B_{ii}(e_i^2 - \hat{\sigma}_i^2)$, we represent the leave-one-out variance estimator as

$$\begin{aligned}
\hat{\sigma}_i^2 &= y_i(y_i - w_i'\hat{\gamma}_{-i}) = y_i M_{W,ii}^{-1}(y_i - w_i'\hat{\gamma}) \\
&= y_i M_{W,ii}^{-1}\hat{e}_i = y_i M_{W,ii}^{-1} \sum_{\ell=1}^n M_{W,i\ell} y_\ell \\
&= y_i M_{W,ii}^{-1} \sum_{\ell=1}^n M_{W,i\ell} \left(x_\ell'\beta + f(z_\ell) + e_\ell\right) \\
&= y_i M_{W,ii}^{-1} \sum_{\ell=1}^n M_{W,i\ell} \left(x_\ell'\beta + f(z_\ell)\right) + y_i M_{W,ii}^{-1} \sum_{\ell=1}^n M_{W,i\ell} e_\ell.
\end{aligned}$$

Thus, the variance is now expressed as:

$$\begin{aligned}
&\sum_{i=1}^n \left(\sum_{\ell=1}^n M_{W,\ell\ell}^{-1} B_{\ell\ell} M_{W,i\ell} \left(x_\ell'\beta + f(z_\ell)\right) \right)^2 \sigma_i^2 + 2\sum_{i=1}^n \sum_{\ell \neq i} M_{W,ii}^{-2} B_{ii}^2 M_{W,i\ell}^2 \sigma_i^2 \sigma_\ell^2 \\
&\qquad \leq \frac{1}{c^2} \max_i \sigma_i^2 \max_i \left(x_i'\beta + f(z_i)\right)^2 \sum_{i=1}^n B_{ii}^2 + \frac{2}{c} \max_i \sigma_i^4 \sum_{i=1}^n B_{ii}^2 = o(1),
\end{aligned}$$

because $\min_i M_{W,ii} \geq c > 0$, $\sum_{i=1}^n B_{ii}^2 \leq \text{trace}(\tilde{A}^2) = o(1)$, and $\max_i \left(x_i'\beta + f(z_i)\right)^2 \leq 2\max_i (x_i'\beta)^2 + 2\max_i f(z_i)^2 = \mathcal{O}(1)$.

Because we have that

$$\mathbb{E}[\hat{\theta} - \theta] \xrightarrow{p} 0, \quad \text{var}[\hat{\theta} - \theta] \xrightarrow{p} 0,$$

the proposed estimator $\hat{\theta}$ is consistent. □

Proof of Lemma 2. We prove the result by considering a second-order approximation of $\hat{\theta}_{\text{JLA}} - \hat{\theta}$ around $\hat{a}_i := (1 - P_{W,ii})^{-1}(\hat{P}_{W,ii} - P_{W,ii})$ as

$$(\hat{\theta}_{\text{JLA}} - \hat{\theta})_2 := \sum_{i=1}^{n} \hat{\sigma}_i^2 \left(B_{ii} - \hat{B}_{ii} - \hat{B}_{ii}\hat{a}_i - \hat{B}_{ii} \left(\hat{a}_i^2 - \frac{1}{m} \frac{3P_{W,ii}^3 + P_{W,ii}^2}{1 - P_{W,ii}} \right) \right),$$

and an approximation error that we show to be negligible,

$$\text{AE}_2 := \sum_{i=1}^{n} \hat{\sigma}_i^2 \hat{B}_{ii} \left(\frac{1}{m} \frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2 - (3P_{W,ii}^3 + P_{W,ii}^2)(1 - \hat{a}_i)^2}{(1 - \hat{a}_i)^2 (1 - P_{W,ii})} - \frac{\hat{a}_i^3}{1 - \hat{a}_i} \right).$$

To decompose $\hat{\theta}_{\text{JLA}} - \hat{\theta} = (\hat{\theta}_{\text{JLA}} - \hat{\theta})_2 + \text{AE}_2$, note that

$$\begin{aligned}
\hat{\sigma}_{i,\text{JLA}}^2 &= \frac{y_i(y_i - w_i'\hat{\gamma})}{1 - \hat{P}_{W,ii}} \left(1 - \frac{1}{m} \frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2}{1 - \hat{P}_{W,ii}} \right) \\
&= \frac{1 - P_{W,ii}}{1 - \hat{P}_{W,ii}} \hat{\sigma}_i^2 \left(1 - \frac{1}{m} \frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2}{1 - \hat{P}_{W,ii}} \right),
\end{aligned}$$

so that

$$\begin{aligned}
\hat{\theta}_{\text{JLA}} - \hat{\theta} &= \sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2 - \hat{B}_{ii}\hat{\sigma}_{i,\text{JLA}}^2 \\
&= \sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2 - \hat{B}_{ii}\hat{\sigma}_i^2 \left(1 - \frac{1}{m} \frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2}{1 - \hat{P}_{W,ii}} \right) \\
&= \sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2 + \hat{B}_{ii}\hat{\sigma}_i^2 \left(\frac{1}{m} \frac{1 - P_{W,ii}}{1 - \hat{P}_{W,ii}} \frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2}{1 - \hat{P}_{W,ii}} - \frac{1 - P_{W,ii}}{1 - \hat{P}_{W,ii}} \right).
\end{aligned}$$

Add and subtract $m^{-1}\hat{\sigma}_i^2 \hat{B}_{ii}(1 - P_{W,ii})^{-1}(3P_{W,ii}^3 + P_{W,ii}^2)$ to obtain

$$\begin{aligned}
\hat{\theta}_{\text{JLA}} - \hat{\theta} &= \sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2 + \frac{1}{m}\hat{\sigma}_i^2 \hat{B}_{ii} \frac{3P_{W,ii}^3 + P_{W,ii}^2}{1 - P_{W,ii}} + \hat{B}_{ii}\hat{\sigma}_i^2 \left(\frac{1}{m} \frac{1 - P_{W,ii}}{1 - \hat{P}_{W,ii}} \frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2}{1 - \hat{P}_{W,ii}} - \frac{1 - P_{W,ii}}{1 - \hat{P}_{W,ii}} \right) \\
&\quad - \frac{1}{m}\hat{\sigma}_i^2 \hat{B}_{ii} \frac{3P_{W,ii}^3 + P_{W,ii}^2}{1 - P_{W,ii}} \\
&= \sum_{i=1}^{n} \hat{\sigma}_i^2 \left(B_{ii} + \frac{1}{m} \hat{B}_{ii} \frac{3P_{W,ii}^3 + P_{W,ii}^2}{1 - P_{W,ii}} \right) \\
&\quad + \sum_{i=1}^{n} \hat{\sigma}_i^2 \hat{B}_{ii} \left(\frac{1}{m} \frac{(1 - P_{W,ii})^2 (3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2) - (1 - \hat{P}_{W,ii})^2 (3P_{W,ii}^3 + P_{W,ii}^2)}{(1 - \hat{P}_{W,ii})^2 (1 - P_{W,ii})} - \frac{1 - P_{W,ii}}{1 - \hat{P}_{W,ii}} \right),
\end{aligned}$$

and using $(1-\hat{P}_{W,ii})^{-1}(1-P_{W,ii}) = (1-\hat{a}_i)^{-1}$, and expanding up to the third order as

$$\begin{aligned}\frac{1-P_{W,ii}}{1-\hat{P}_{W,ii}} &= 1 + \frac{1-P_{W,ii}}{1-\hat{P}_{W,ii}}\hat{a}_i \\ &= 1 + \hat{a}_i + \hat{a}_i^2 + \frac{1-P_{W,ii}}{1-\hat{P}_{W,ii}}\hat{a}_i^3,\end{aligned}$$

we have that

$$\begin{aligned}\hat{\theta}_{\mathrm{JLA}} - \hat{\theta} &= \sum_{i=1}^n \hat{\sigma}_i^2 \left(B_{ii} + \frac{1}{m}\hat{B}_{ii}\frac{3P_{W,ii}^3 + P_{W,ii}^2}{1-P_{W,ii}} \right) \\ &\quad + \sum_{i=1}^n \hat{\sigma}_i^2 \hat{B}_{ii} \left(\frac{1}{m}\frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2 - (1-\hat{a}_i)^2(3P_{W,ii}^3 + P_{W,ii}^2)}{(1-\hat{a}_i)^2(1-P_{W,ii})} - \left(1 + \hat{a}_i + \hat{a}_i^2 + \frac{\hat{a}_i^3}{1-\hat{a}_i}\right)\right) \\ &= (\hat{\theta}_{\mathrm{JLA}} - \hat{\theta})_2 + \mathrm{AE}_2.\end{aligned}$$

To describe the bias, we note that $\hat{P}_{W,ii}$, $\hat{B}_{ii}$, and $\hat{\sigma}_i^2$ are independent of each other, $\mathbb{E}[\hat{P}_{W,ii}] = P_{W,ii}$, $\mathbb{E}[\hat{B}_{ii}] = B_{ii}$, $\mathbb{E}[\hat{\sigma}_i^2] = \sigma_i^2 + \mathcal{O}_p(\sqrt{n}/k^{\alpha_f}\sqrt{p})$, and using properties of the Rademacher random variables,

$$\mathrm{var}[\hat{a}_i] = \frac{2}{m}\frac{P_{W,ii} - \sum_{\ell=1}^n P_{W,i\ell}^4}{(1-P_{W,ii})^2} = \frac{1}{m}\frac{3P_{W,ii}^3 + P_{W,ii}^2}{1-P_{W,ii}} + \frac{P_{W,ii}(1-P_{W,ii})^2 - 2\sum_{\ell\neq i}^n P_{W,i\ell}^4}{m(1-P_{W,ii})^2}.$$

Therefore, in total we have:

$$\mathbb{E}[(\hat{\theta}_{\mathrm{JLA}} - \hat{\theta})_2] = -\sum_{i=1}^n \sigma_i^2 B_{ii}\left(\mathrm{var}[\hat{a}_i] - \frac{1}{m}\frac{3P_{W,ii}^3 + P_{W,ii}^2}{1-P_{W,ii}}\right) + \mathcal{O}_p(\sqrt{n}/k^{\alpha_f}\sqrt{p}),$$

or, assuming $k \to \infty$,

$$\mathbb{E}[(\hat{\theta}_{\mathrm{JLA}} - \hat{\theta})_2] = \mathrm{B}_m + o(1), \quad \mathrm{B}_m := \sum_{i=1}^n B_{ii}\sigma_i^2 \left(\frac{2\sum_{\ell\neq i} P_{W,i\ell}^4 - P_{W,ii}^2(1-P_{W,ii})^2}{m(1-P_{W,ii})^2}\right).$$

Focusing on the variance next, denote $y := (y_1, \ldots, y_n)'$, so that

$$\begin{aligned}\mathrm{var}\left[\sum_{i=1}^n \hat{\sigma}_i^2(B_{ii} - \hat{B}_{ii})\right] &= \mathbb{E}\left[\mathrm{var}\left[\sum_{i=1}^n \hat{\sigma}_i^2\hat{B}_{ii}\right]\middle| y\right] + \mathrm{var}\left[\mathbb{E}\left[\sum_{i=1}^n \hat{\sigma}_i^2\hat{B}_{ii}\right]\middle| y\right] = \mathbb{E}\left[\mathrm{var}\left[\sum_{i=1}^n \hat{\sigma}_i^2\hat{B}_{ii}\right]\middle| y\right] \\ &\leq 2m^{-1}\sum_{i=1}^n\sum_{\ell=1}^n B_{i\ell}^2\mathbb{E}[\hat{\sigma}_i^2\hat{\sigma}_\ell^2] = \mathcal{O}\left(m^{-1}\mathrm{trace}(\tilde{A}^2)\right),\end{aligned}$$

$$\begin{aligned}\mathrm{var}\left[\sum_{i=1}^n \hat{\sigma}_i^2\hat{B}_{ii}\hat{a}_i\right] &= \mathbb{E}\left[\mathrm{var}\left[\sum_{i=1}^n \hat{\sigma}_i^2\hat{B}_{ii}\hat{a}_i\middle| y, R_B\right]\right] \leq 2m^{-1}\sum_{i=1}^n\sum_{\ell=1}^n P_{W,i\ell}^2\frac{\mathbb{E}[\hat{B}_{ii}\hat{B}_{\ell\ell}]\mathbb{E}[\hat{\sigma}_i^2\hat{\sigma}_\ell^2]}{(1-P_{W,ii})(1-P_{W,\ell\ell})} \\ &= \mathcal{O}\left(m^{-1}\mathrm{trace}(\tilde{A}^2) + m^{-2}\mathrm{trace}(\tilde{A}_1^2)^{1/2}\mathrm{trace}(\tilde{A}_2^2)^{1/2}\right)\end{aligned}$$

for $\tilde{A}_k := S_{xx}^{-1/2} A_k' A_k S_{xx}^{-1/2}$ for $k = 1, 2$. Regarding the ensuing terms, it holds that:

$$\begin{aligned}\operatorname{var}\left[\sum_{i=1}^{n} \hat{\sigma}_i^2 \hat{B}_{ii} \left(\hat{a}_i^2 - \operatorname{var}[\hat{a}_i]\right)\right] &= \sum_{i=1}^{n}\sum_{\ell=1}^{n} \mathbb{E}[\hat{B}_{ii}\hat{B}_{\ell\ell}]\mathbb{E}[\hat{\sigma}_i^2\hat{\sigma}_\ell^2] \operatorname{cov}[\hat{a}_i^2, \hat{a}_\ell^2] \\ &= \mathcal{O}\left(m^{-2}\operatorname{trace}(\tilde{A}^2) + m^{-3}\operatorname{trace}(\tilde{A}_1^2)^{1/2}\operatorname{trace}(\tilde{A}_2^2)^{1/2}\right),\end{aligned}$$

$$\operatorname{var}\left[\sum_{i=1}^{n} \hat{\sigma}_i^2 (\hat{B}_{ii} - B_{ii}) \frac{2\sum_{\ell\neq i}^{n} P_{W,i\ell}^4 - P_{W,ii}(1-P_{W,ii})^2}{m(1-P_{W,ii})^2}\right] = \mathcal{O}\left(m^{-3}\operatorname{trace}(\tilde{A}^2)\right),$$

$$\operatorname{var}\left[\sum_{i=1}^{n} B_{ii} \left(\hat{\sigma}_i^2 - \sigma_i^2\right) \frac{2\sum_{\ell\neq i}^{n} P_{W,i\ell}^4 - P_{W,ii}(1-P_{W,ii})^2}{m(1-P_{W,ii})^2}\right] = \mathcal{O}\left(m^{-2}\operatorname{var}[\hat{\theta}]\right).$$

Because $\operatorname{trace}(\tilde{A}^2) = \mathcal{O}(\operatorname{var}[\hat{\theta}])$ and $m^{-4}\operatorname{var}[\hat{\theta}]^{-2}\operatorname{var}[\hat{\theta}_1]\operatorname{var}[\hat{\theta}_2] = o(1)$, it can be ultimately established that $\operatorname{var}[\hat{\theta}]^{-1/2}\left((\hat{\theta}_{\mathrm{JLA}} - \hat{\theta})_2 - \mathrm{B}_m\right) = o_p(1)$.

Using that $\mathbb{E}[\hat{a}_i^3] = \mathcal{O}(m^{-2})$, $\mathbb{E}[\hat{a}_i^4] = \mathcal{O}(m^{-2})$, and $\max_i |\hat{a}_i| = o_p(\log n/\sqrt{m})$, the terms in the approximation error are as follows:

$$\begin{aligned}\sum_{i=1}^{n} \hat{\sigma}_i^2 \hat{B}_{ii}\hat{a}_i^3 + \sum_{i=1}^{n} \hat{\sigma}_i^2 \hat{B}_{ii}\hat{a}_i^4 &= m^{-2}\mathcal{O}_p\left(\mathbb{E}[\hat{\theta}_{1,\mathrm{PI}} - \theta_1] + \mathbb{E}[\hat{\theta}_{2,\mathrm{PI}} - \theta_2]\right), \\ \sum_{i=1}^{n} \hat{\sigma}_i^2 \hat{B}_{ii} \frac{\hat{a}_i^5}{1-\hat{a}_i} &= \frac{\log n}{m^{5/4}}\mathcal{O}_p\left(\mathbb{E}[\hat{\theta}_{1,\mathrm{PI}} - \theta_1] + \mathbb{E}[\hat{\theta}_{2,\mathrm{PI}} - \theta_2]\right), \\ \frac{1}{m}\sum_{i=1}^{n} \hat{\sigma}_i^2 \hat{B}_{ii} &\frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2 - (3P_{W,ii}^3 + P_{W,ii}^2)(1-\hat{a}_i)^2}{(1-\hat{a}_i)^2(1-P_{W,ii})} \\ &= \left(m^{-2} + \frac{\log n}{p^{5/4}}\right)\mathcal{O}_p\left(\mathbb{E}[\hat{\theta}_{1,\mathrm{PI}} - \theta_1] + \mathbb{E}[\hat{\theta}_{2,\mathrm{PI}} - \theta_2]\right).\end{aligned}$$

□

Proof of Theorem 3. Representation in the theorem holds because

$$\sum_{\ell=1}^{r} \lambda_\ell \hat{b}_\ell^2 = \hat{\beta}' S_{xx}^{1/2} Q D Q' S_{xx}^{1/2} \hat{\beta} = \hat{\beta}' S_{xx}^{1/2} S_{xx}^{-1/2} A S_{xx}^{-1/2} S_{xx}^{1/2} \hat{\beta} = \hat{\beta}' A \hat{\beta},$$

and

$$\sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2 = \operatorname{trace}(A\widehat{\operatorname{var}}[\hat{\beta}]) = \operatorname{trace}(D\widehat{\operatorname{var}}[\hat{b}]) = \sum_{\ell=1}^{r} \lambda_\ell \widehat{\operatorname{var}}[\hat{b}_\ell].$$

We prove the theorem in three steps.

Approximation. Equivalently, represent $\hat{\theta}$ as

$$\hat{\theta} = \sum_{\ell=1}^{r} \lambda_\ell \left(\hat{b}_\ell^2 - \operatorname{var}[\hat{b}_\ell]\right) + \sum_{i=1}^{n} B_{ii}(\sigma_i^2 - \hat{\sigma}_i^2),$$

and below we show that the second term is asymptotically dominated by the variance of the estimator, $\operatorname{var}[\hat{\theta}]$. Given that, the asymptotic distribution of $\hat{\theta}$ is then driven by the joint distribution of random vector $\hat{b}$.

We need to show that the second term is asymptotically mean-zero and is of smaller order than variance of $\hat{\theta}$. The first claim is immediate because we have shown in Lemma 1 that

$$\mathbb{E}[\hat{\sigma}_i^2 - \sigma_i^2] = \mathcal{O}_p(\sqrt{n}/k^{\alpha_f}\sqrt{p}) \xrightarrow{p} 0,$$

so that

$$\mathbb{E}\left[\sum_{i=1}^n B_{ii}(\sigma_i^2 - \hat{\sigma}_i^2)\right] \xrightarrow{p} 0.$$

Now, rewrite

$$\begin{aligned}\sum_{i=1}^n B_{ii}(\hat{\sigma}_i^2 - \sigma_i^2) &= \sum_{i=1}^n B_{ii} M_{W,ii}^{-1} x_i'\beta \sum_{\ell=1}^n M_{W,i\ell} e_\ell + \sum_{i=1}^n (e_i^2 - \sigma_i^2) \\ &\quad + \sum_{i=1}^n B_{ii} M_{W,ii}^{-1} f(z_i) \sum_{\ell=1}^n M_{W,i\ell} e_\ell + \sum_{i=1}^n B_{ii} M_{W,ii}^{-1} \sum_{\ell \neq i} M_{W,i\ell} e_i e_\ell.\end{aligned}$$

The variances of the first and the third terms are

$$\sum_{\ell=1}^n \sigma_\ell^2 \left(\sum_{i=1}^n M_{W,i\ell} B_{ii} M_{W,ii}^{-1} x_i'\beta\right)^2 \leq \max_i \sigma_i^2 \sum_{i=1}^n B_{ii}^2 M_{W,ii}^{-2} (x_i'\beta)^2 \leq \max_i \sigma_i^2 \max_i (x_i'\beta)^2 M_{W,ii}^{-2} \sum_{i=1}^n B_{ii}^2,$$

$$\sum_{\ell=1}^n \sigma_\ell^2 \left(\sum_{i=1}^n M_{W,i\ell} B_{ii} M_{W,ii}^{-1} f(z_i)\right)^2 \leq \max_i \sigma_i^2 \sum_{i=1}^n B_{ii}^2 M_{W,ii}^{-2} f(z_i)^2 \leq \max_i \sigma_i^2 \max_i f(z_i)^2 M_{W,ii}^{-2} \sum_{i=1}^n B_{ii}^2,$$

and of the second and the forth

$$\sum_{i=1}^n B_{ii}^2 \operatorname{var}[e_i^2] \leq \max_i \mathbb{E}[e_i^4] \sum_{i=1}^n B_{ii}^2,$$

$$\sum_{i=1}^n \sum_{\ell \neq i} (B_{ii}^2 M_{W,ii}^{-2} + B_{ii} M_{W,ii}^{-1} B_{\ell\ell} M_{W,\ell\ell}^{-1}) M_{W,i\ell}^2 \sigma_i^2 \sigma_\ell^2 \leq 2 \max_i \sigma_i^4 M_{W,ii}^{-2} \sum_{i=1}^n B_{ii}^2.$$

Because each variance is bounded by $C \sum_{i=1}^n B_{ii}^2$, to show that it is of smaller order than the variance of $\hat{\theta}$, we need $\operatorname{var}[\hat{\theta}]^{-1} \sum_{i=1}^n B_{ii}^2 = o(1)$. It holds because

$$\operatorname{var}[\hat{\theta}]^{-1} \sum_{i=1}^n B_{ii}^2 \leq \max_i v_i' v_i \operatorname{var}[\hat{\theta}]^{-1} \sum_{\ell=1}^r \lambda_\ell^2 \leq \max_i v_i' v_i \max_i \sigma_i^{-4} = o(1).$$

Variance estimator consistency. Now we show that the variance estimator is consistent, i.e. $\operatorname{var}[\hat{b}]^{-1} \widehat{\operatorname{var}}[\hat{b}] \xrightarrow{p} I_r$. For it we need to show that

$$\operatorname{var}[\vartheta'\hat{b}]^{-1} \left(\widehat{\operatorname{var}}[\vartheta'\hat{b}] - \operatorname{var}[\vartheta'\hat{b}]\right) = o_p(1), \quad \vartheta \in \mathbb{R}^r, \quad \vartheta'\vartheta = 1$$

for some non-random ϑ. Rewrite the expression above as

$$\delta(\vartheta) := \sum_{i=1}^n v_i(\vartheta)(\hat{\sigma}_i^2 - \sigma_i^2), \tag{8}$$

where

$$v_i(\vartheta) := \frac{(\vartheta' v_i)^2}{\sum_{i=1}^n \sigma_i^2 (\vartheta' v_i)^2}.$$

We know that $\mathbb{E}[\delta(\vartheta)] = o_p(1)$ because by the triangle inequality, the Cauchy-Schwartz inequality, and $|\mathbb{E}[\hat{\sigma}_i^2 - \sigma_i^2]| = o_p(1)$,

$$|v(\vartheta)\mathbb{E}[\hat{\sigma}_i^2 - \sigma_i^2]| \leq (v(\vartheta)^2)^{1/2} \cdot \left(|\mathbb{E}[\hat{\sigma}_i^2 - \sigma_i^2]|^2\right)^{1/2} = o_p(1)$$

for $i = 1, \ldots, n$. The variance of $\delta(\vartheta)$ is

$$\begin{aligned}
\sum_{i=1}^n \delta(\vartheta)^2 \operatorname{var}[\hat{\sigma}_i^2] &\leq \sum_{i=1}^n v_i(\vartheta)^4 \\
&\leq \max_i \sigma_i^{-4} \max_i v_i v_i' \frac{\vartheta'\vartheta}{\sum_{i=1}^n v_i v_i' \vartheta'\vartheta} \\
&= \max_i \sigma_i^{-4} \max_i v_i v_i' = o(1)
\end{aligned}$$

because $\max_i v_i v_i' = o(1)$ by assumption.

Asymptotic normality. Our objective is to prove that

$$\operatorname{var}[\vartheta'\hat{b}]^{-1/2} \left(\vartheta'\hat{b} - \vartheta' b\right) \xrightarrow{d} \mathcal{N}(0,1),$$

where $\hat{b} := Q' S_{xx}^{1/2} \hat{\beta}$, and $b := Q' S_{xx}^{1/2} \beta$. Lyapunov's condition implies that it is sufficient to show that

$$\operatorname{var}[\vartheta'\hat{b}]^{-2} \sum_{i=1}^n \mathbb{E}\left[\left(\vartheta'(\hat{b} - b)\right)^4\right] = o_p(1).$$

Because we have that

$$\begin{aligned}
\vartheta'(\hat{b} - b) &= \vartheta'(Q' S_{xx}^{1/2} \hat{\beta} - Q' S_{xx}^{1/2} \beta) \\
&= \sum_{i=1}^n \vartheta' v_i \left(f(z_i) + e_i\right),
\end{aligned}$$

and $(a+b)^4 \leq C(a^4 + b^4)$ for some constant C, Lyapunov's condition is equivalent to

$$\operatorname{var}[\vartheta'\hat{b}]^{-2} \sum_{i=1}^n C\left(f(z_i)^4 + \mathbb{E}[e_i^4]\right) \cdot (\vartheta' v_i)^4 = o_p(1).$$

It holds because $\max_i |f(z_i)| = \mathcal{O}(1)$, and $\max_i \mathbb{E}[e_i^4] = \mathcal{O}(1)$ by assumption, so that $\max_i f(z_i)^4 + \mathbb{E}[e_i^4] = \mathcal{O}(1)$, also $\max_i (\vartheta' v_i)^2 \leq \max_i v_i' v_i = o(1)$, $\sum_{i=1}^n (\vartheta' v_i)^2 = \sum_{i=1}^n \vartheta' v_i v_i' \vartheta = 1$, and $\operatorname{var}[\vartheta'\hat{b}]^{-2} \leq \max_i \sigma_i^{-2} = \mathcal{O}(1)$. □

Proof of Theorem 4. We derive the limiting distribution of $\hat{\theta}$ with growing rank based on the following result regarding the joint normality of independent and not necessarily identical random variables as in Kline, Saggio, and Sølvsten (2020).

Let $\{q_{n,i}\}_{i,n}$ be a triangular array of row-wise independent random variables with

$\mathbb{E}[q_{n,i}] = 0$ and $\text{var}[q_{n,i}] = \sigma^2_{n,i}$, let $\{\dot{w}_{n,i}\}_{i,n}$ be a triangular array of non-random weights that satisfy $\sum_{i=1}^n \dot{w}_{n,i}\sigma^2_{n,i} = 1$ for $\forall n$, and let $(Q_n)_n$ be a sequence of symmetric non-random matrices in $\mathbb{R}^{n\times n}$ with zeros on the diagonal and having $2\sum_{i=1}^n \sum_{\ell\neq i} Q^2_{n,i\ell}\sigma^2_{n,i}\sigma^2_{n,\ell} = 1$. Define

$$\mathcal{S}_n := \sum_{i=1}^n \dot{w}_{n,i} q_{n,i}, \quad \mathcal{U}_n := \sum_{i=1}^n \sum_{\ell\neq i} Q_{n,i\ell} q_{n,i} q_{n,\ell}.$$

Lemma 3. *If* $\max_i \mathbb{E}[q^4_{n,i}] + \sigma^{-2}_{n,i} = O(1)$, *(i)* $\max_i \dot{w}^2_{n,i} = o(1)$, *and (ii)* $trace(Q^4_n) = o(1)$, *then* $(\mathcal{S}_n, \mathcal{U}_n)' \xrightarrow{d} \mathcal{N}(0, I_2)$.

Proof. See Appendix B in Kline, Saggio, and Sølvsten (2020) and Appendix A2 in Solvsten (2020). □

The U-statistic representation holds because

$$\begin{aligned}
\hat{\theta} &= \sum_{i=1}^n y_i \tilde{w}_i' \hat{\gamma}_{-i} = \sum_{i=1}^n y_i \tilde{w}_i' \left(S_{ww}^{-1} - w_i w_i'\right) \sum_{\ell\neq i} w_\ell y_\ell \\
&= \sum_{i=1}^n y_i \tilde{w}_i' \left(S_{ww}^{-1} + \frac{S_{ww}^{-1} w_i w_i' S_{ww}^{-1}}{1 - w_i' S_{ww}^{-1} w_i}\right) \sum_{\ell\neq i} w_\ell y_\ell \\
&= \sum_{i=1}^n y_i \tilde{w}_i' \left(S_{ww}^{-1} + M_{W,ii}^{-1} S_{ww}^{-1} w_i w_i' S_{ww}^{-1}\right) \sum_{\ell\neq i} w_\ell y_\ell \\
&= \sum_{i=1}^n y_i w_i' S_{ww}^{-1} \check{A} S_{ww}^{-1} \sum_{\ell\neq i} w_\ell y_\ell + \sum_{i=1}^n y_i w_i' S_{ww}^{-1} \check{A} M_{W,ii}^{-1} S_{ww}^{-1} w_i w_i' S_{ww}^{-1} \sum_{\ell\neq i} w_\ell y_\ell \\
&= \sum_{i=1}^n \sum_{\ell\neq i} y_i y_\ell w_i' S_{ww}^{-1} \check{A} S_{ww}^{-1} w_\ell + \sum_{i=1}^n \sum_{\ell\neq i} y_i y_\ell M_{W,ii}^{-1} w_i' S_{ww}^{-1} \check{A} S_{ww}^{-1} w_i w_i' S_{ww}^{-1} w_\ell \\
&= \sum_{i=1}^n \sum_{\ell\neq i} y_i y_\ell B_{W,i\ell} + y_i y_\ell M_{W,ii}^{-1} B_{W,ii}(1 - M_{W,i\ell}) = \sum_{i=1}^n \sum_{\ell\neq i} C_{i\ell} y_i y_\ell.
\end{aligned}$$

Now, define $\check{e}_i := f(z_i) - p_k(z_i)'\alpha$ to be an approximation error (it is implicitly indexed by the unknown function but we omit this dependence for brevity). We can write the difference

$$\begin{aligned}
\hat{\theta} - \theta &= \sum_{i=1}^n \sum_{\ell\neq i} C_{i\ell} y_i y_\ell - \sum_{i=1}^n \gamma' w_i \tilde{w}_i' \gamma \\
&= \sum_{i=1}^n \sum_{\ell\neq i} C_{i\ell} \left(x_i'\beta + f(z_i) + e_i\right)\left(x_\ell'\beta + f(z_\ell) + e_\ell\right) - \sum_{i=1}^n \gamma' w_i \tilde{w}_i' \gamma \\
&= \sum_{i=1}^n \sum_{\ell\neq i} C_{i\ell} \left(\gamma' w_i + e_i + \check{e}_i\right)\left(\gamma' w_\ell + e_\ell + \check{e}_\ell\right) - \sum_{i=1}^n \gamma' w_i \tilde{w}_i' \gamma \\
&= \sum_{i=1}^n \sum_{\ell\neq i} C_{i\ell} \gamma' w_i \gamma' w_\ell + \sum_{i=1}^n \sum_{\ell\neq i} C_{i\ell}(\gamma' w_i e_\ell + \gamma' w_\ell e_i) - \sum_{i=1}^n \gamma' w_i \tilde{w}_i' \gamma \\
&\quad + \sum_{i=1}^n \sum_{\ell\neq i} C_{i\ell}(\gamma' w_i \check{e}_\ell + \gamma' w_\ell \check{e}_i + e_i \check{e}_\ell + e_\ell \check{e}_i) + \sum_{i=1}^n \sum_{\ell\neq i} C_{i\ell} e_i e_\ell + \sum_{i=1}^n \sum_{\ell\neq i} C_{i\ell} \check{e}_i \check{e}_\ell
\end{aligned}$$

$$
\begin{aligned}
&= \sum_{i=1}^{n} e_i \sum_{\ell \neq i} (\gamma' w_i + \gamma' w_\ell) C_{i\ell} + \sum_{i=1}^{n} \check{e}_i \sum_{\ell \neq i} (\gamma' w_i + \gamma' w_\ell + e_i + e_\ell) \\
&\quad + \sum_{i=1}^{n} \sum_{\ell \neq i} C_{i\ell} e_i e_\ell + \sum_{i=1}^{n} \sum_{\ell \neq i} C_{i\ell} \check{e}_i \check{e}_\ell \\
&= \sum_{i=1}^{n} (2\tilde{w}_i' \gamma - \check{w}_i' \gamma) e_i + \sum_{i=1}^{n} \sum_{\ell \neq i} C_{i\ell} e_i e_\ell + \mathcal{O}_p(nk^{-\alpha_f}),
\end{aligned}
$$

where the last equality follows from defining $\check{w}_i := \sum_{\ell=1}^{n} M_{W,i\ell} \frac{B_{W,\ell\ell}}{1-P_{W,\ell\ell}} w_\ell$, and the fact that by the Assumption 1

$$
\check{e}_i^2 \leq \left(f(z_i) - p_k(z_i)'\alpha\right)^2 = \mathbb{E}[\left(f(z_i) - p_k(z_i)'\alpha\right)^2] \leq \min_{\alpha \in \mathbb{R}^k} \mathbb{E}[\left(f(z_i) - p_k(z_i)'\alpha\right)^2] \leq Ck^{-2\alpha_f}.
$$

As $k \to \infty$, we have that

$$
\hat{\theta} - \theta = \sum_{i=1}^{n} \left(2\tilde{w}_i' \gamma - \check{w}_i' \gamma\right) e_i + \sum_{i=1}^{n} \sum_{\ell \neq i} C_{i\ell} e_i e_\ell + o_p(1).
$$

Having dispensed with asymptotically negligible contributions to $\hat{\theta}$, asymptotic variance is

$$
\mathrm{var}[\hat{\theta}] = \sum_{i=1}^{n} (2\tilde{w}_i' \gamma - \check{w}_i' \gamma)^2 \sigma_i^2 + 2 \sum_{i=1}^{n} \sum_{\ell \neq i} C_{i\ell} \sigma_i^2 \sigma_\ell^2,
$$

that is, a sum of two components given by

$$
\mathcal{V}_s := \sum_{i=1}^{n} (2\tilde{w}_i' \gamma - \check{w}_i' \gamma)^2 \sigma_i^2, \quad \mathcal{V}_u := 2 \sum_{i=1}^{n} \sum_{\ell \neq i} C_{i\ell} \sigma_i^2 \sigma_\ell^2.
$$

The normalized difference is then given by

$$
\mathrm{var}[\hat{\theta}]^{-1/2} (\hat{\theta} - \theta) = \omega_1 \mathcal{S}_n + \omega_2 \mathcal{U}_n,
$$

with $\omega_1 := \mathrm{var}[\hat{\theta}]^{-1/2} \mathcal{V}_s^{1/2}$, $\omega_2 := \mathrm{var}[\hat{\theta}]^{-1/2} \mathcal{V}_u^{1/2}$, and

$$
\mathcal{S}_n := \mathcal{V}_s^{-1/2} \sum_{i=1}^{n} (2\tilde{w}_i' \gamma - \check{w}_i' \gamma) e_i, \quad \mathcal{U}_n := \mathcal{V}_u^{-1/2} \sum_{i=1}^{n} \sum_{\ell \neq i} C_{i\ell} e_i e_\ell.
$$

Consider the case where the limit of ω_1 is nonzero. If it is not, then asymptotic normality of the difference $\mathrm{var}[\hat{\theta}]^{-1/2}(\hat{\theta} - \theta)$ is implied by asymptotic normality of $\mathcal{U}_n$. Using notation of Lemma 3, we have that $\dot{w}_i = \mathcal{V}_s^{-1/2}(2\tilde{w}_i' \gamma - \check{w}_i' \gamma)$, and $Q_{i\ell} = \mathcal{V}_u^{-1/2} C_{i\ell}$.

To verify the condition *(i)* of Lemma 3, note that

$$
\begin{aligned}
\max_i \dot{w}_i^2 = \max_i \mathcal{V}_s^{-1} (2\tilde{w}_i' \gamma - \check{w}_i' \gamma)^2 &\leq \max_i 4\mathcal{V}_s^{-1} \left((\tilde{w}_i' \gamma)^2 + (\check{w}_i' \gamma)^2\right) \\
&= \max_i 4\omega_1^{-2} \frac{(\tilde{w}_i' \gamma)^2 + (\check{w}_i' \gamma)^2}{\mathrm{var}[\hat{\theta}]} = o(1),
\end{aligned}
$$

where the last equality follows from Theorem 4 *(i)*, and the nonzero limit of ω_1.

For the condition *(ii)* of Lemma 3, denote $\tilde{A}_W := S_{ww}^{-1/2}\check{A}S_{ww}^{-1/2}$, and note that the first r eigenvalues of the $\tilde{A}_W$ matrix are equal to eigenvalues of the $\tilde{A}$ matrix. Now, with constants c_U and c_L not dependent on n, we have that $\text{trace}(C^4) \leq c_U \cdot \text{trace}(B_W^4) = c_U \cdot \text{trace}(\tilde{A}_W^4) \leq c_U\lambda_1^2 \cdot \text{trace}(\tilde{A}_W^2)$, and $\mathcal{V}_u \geq c_L \min_i \sigma_i^4 \cdot \text{trace}(\tilde{A}_W)$, which implies

$$\text{trace}(Q^4) \leq \frac{c_U\lambda_1^2 \cdot \text{trace}(\tilde{A}_W^2)}{\left(c_L \min_i \sigma_i^4 \cdot \text{trace}(\tilde{A}_W^2)\right)^2} = \mathcal{O}\left(\frac{\lambda_1^2}{\text{trace}(\tilde{A}_W^2)}\right) = o(1),$$

where the last equality follows from Theorem 4 *(ii)*. □

Sherman-Morrison representation. The representation in (4) holds due to application of the Sherman-Morrison formula to the leave-one-out estimator of γ. Denoting $S_{ww} := \sum_{i=1}^n w_i w_i'$, it holds that

$$\left(S_{ww} - w_i w_i'\right)^{-1} = S_{ww}^{-1} + \frac{S_{ww}^{-1} w_i w_i' S_{ww}^{-1}}{1 - w_i' S_{ww}^{-1} w_i}.$$

Plugging the above into the definition of $\hat{\gamma}_{-i}$, we obtain

$$\begin{aligned}
\hat{\gamma}_{-i} &= \left(S_{ww}^{-1} + \frac{S_{ww}^{-1} w_i w_i' S_{ww}^{-1}}{1 - w_i' S_{ww}^{-1} w_i}\right)\left(\sum_{i=1}^n w_i y_i - w_i y_i\right) \\
&= S_{ww}^{-1}\sum_{i=1}^n w_i y_i - S_{ww} w_i y_i + \frac{S_{ww}^{-1} w_i w_i' S_{ww}^{-1} \sum_{i=1}^n w_i y_i}{1 - w_i' S_{ww}^{-1} w_i} - \frac{S_{ww}^{-1} w_i w_i' S_{ww}^{-1} w_i y_i}{1 - w_i' S_{ww}^{-1} w_i} \\
&= \hat{\gamma} + S_{ww}^{-1} w_i \left(\frac{w_i'\hat{\gamma}}{1 - w_i' S_{ww}^{-1} w_i} - y_i - \frac{w_i' S_{ww}^{-1} w_i y_i}{1 - w_i' S_{ww}^{-1} w_i}\right) \\
&= \hat{\gamma} + S_{ww}^{-1} w_i \left(\frac{w_i'\hat{\gamma} - y_i}{1 - w_i' S_{ww}^{-1} w_i}\right) \\
&= \hat{\gamma} - S_{ww}^{-1} w_i \frac{(y_i - w_i'\hat{\gamma})}{1 - w_i' S_{ww}^{-1} w_i}.
\end{aligned}$$

Substituting for $\hat{\gamma}_{-i}$ in (3), we have

$$\begin{aligned}
\hat{\sigma}_i^2 &= y_i\left(y_i - w_i'\hat{\gamma}_{-i}\right) \\
&= y_i\left(y_i - w_i'\left(\hat{\gamma} - S_{ww}^{-1} w_i \frac{(y_i - w_i'\hat{\gamma})}{1 - w_i' S_{ww}^{-1} w_i}\right)\right) \\
&= y_i\left(y_i - w_i'\hat{\gamma} + w_i' S_{ww}^{-1} w_i \frac{(y_i - w_i'\hat{\gamma})}{1 - w_i' S_{ww}^{-1} w_i}\right) \\
&= y_i\left((y_i - w_i'\hat{\gamma})\left(\frac{1}{1 - w_i' S_{ww}^{-1} w_i}\right)\right) \\
&= \frac{y_i \hat{e}_i}{M_{W,ii}}.
\end{aligned}$$

B. Additional Tables

Table 6: Descriptive statistics of the cleaned worker–firm panel

	Mean	SD	p25	Median	p75
A. Worker-year panel					
Age	39.20	10.52	31.00	38.00	47.00
Months worked	8.19	1.69	8.00	9.00	9.00
Tenure (months)	103.57	103.76	22.00	69.00	153.00
Annual wage (k€)	13.92	10.46	7.99	10.53	16.02
Hourly wage (€)	7.17	5.17	4.15	5.33	7.99
Number of jobs	1.01	0.12	1.00	1.00	1.00
B. Worker composition and mobility					
Job-to-job moves (count)	1,803,534	–	–	–	–
Job-to-job movers (%)	9.85%	–	–	–	–
Education: At most primary (%)	47.91%	–	–	–	–
Education: Secondary (%)	34.25%	–	–	–	–
Education: At least bachelor's (%)	17.52%	–	–	–	–
Education missing (%)	0.32%	–	–	–	–
Qualification: Specialized workers (%)	60.44%	–	–	–	–
Qualification: Generic workers (%)	15.27%	–	–	–	–
Qualification missing (%)	11.88%	–	–	–	–
Qualification: Top managers (%)	7.10%	–	–	–	–
Qualification: Middle managers (%)	5.31%	–	–	–	–
Lisboa (%)	35.13%	–	–	–	–
Norte (%)	35.06%	–	–	–	–
Centro (%)	18.28%	–	–	–	–
Alentejo (%)	4.54%	–	–	–	–
Algarve (%)	3.67%	–	–	–	–
Region (NUT2) other (%)	3.32%	–	–	–	–
C. Firm-year panel					
Revenue (M€)	1.64	31.48	0.08	0.20	0.56
Assets (M€)	2.34	73.33	0.07	0.20	0.61
Workers	13.35	119.97	3.00	4.00	9.00
D. Panel counts					
Unique workers	3,532,497	–	–	–	–
Unique firms	484,704	–	–	–	–
Worker-firm observations	18,314,740	–	–	–	–
E. Estimation variable coverage (all runs)					
Missing: Age (%)	0.00%	–	–	–	–
Missing: Assets (%)	8.26%	–	–	–	–
Missing: Education (%)	0.32%	–	–	–	–
Missing: Firm ID (%)	0.00%	–	–	–	–
Missing: Log hourly wage (%)	0.01%	–	–	–	–
Missing: Qualification (%)	11.88%	–	–	–	–
Missing: Revenue (%)	8.26%	–	–	–	–
Missing: Gender (%)	0.00%	–	–	–	–
Missing: Worker ID (%)	0.00%	–	–	–	–
Missing: Year (%)	0.00%	–	–	–	–

Note: Descriptive moments are computed on the cleaned analysis sample (2008–2018). Panels A and C report distributional statistics for worker-year and firm-year variables, respectively. Panels B and D report composition, mobility, and sample-size counts. Percentages are sample shares in the corresponding panel.

Table 7: Coverage of the largest connected component

	Full sample	Largest CC	Share (%)
Observations	14,725,548	13,769,217	93.51
Workers	2,954,762	2,723,884	92.19
Firms	340,929	226,141	66.33
Mean obs per worker	4.98	5.05	–
Mean spells per worker	1.40	1.43	–
B. Coverage for variables used across all runs			
Non-missing: Age	14,725,548	13,769,217	93.51
Non-missing: Assets	14,725,548	13,769,217	93.51
Non-missing: Education	14,725,548	13,769,217	93.51
Non-missing: Firm ID	14,725,548	13,769,217	93.51
Non-missing: Log hourly wage	14,725,548	13,769,217	93.51
Non-missing: Qualification	14,725,548	13,769,217	93.51
Non-missing: Revenue	14,725,548	13,769,217	93.51
Non-missing: Gender	14,725,548	13,769,217	93.51
Non-missing: Worker ID	14,725,548	13,769,217	93.51
Non-missing: Year	14,725,548	13,769,217	93.51

Note: The table compares the full cleaned estimation sample with the largest connected component used for fixed-effects estimation. "Share (%)" is the fraction retained in the largest connected component relative to the full sample for each count measure.

Table 8: Diagnostics for the empirical specification ladder

Model	N	CC N	Workers	Firms	Cont. vars	Groups	k
Panel A: Parsimonious controls							
AKM baseline	18312694	17256654	3282116	333180	0	0	0
Linear additive controls	18312694	17256654	3282116	333180	1	0	2
Heterogeneous linear controls	18312694	17256654	3282116	333180	1	2	3
Nonlinear homogeneous controls	18312694	17256654	3282116	333180	1	0	4
Full model	18312694	17256654	3282116	333180	1	2	9
Panel B: Extended controls							
AKM baseline	14725548	13769217	2723884	226141	0	0	0
Linear additive controls	14725548	13769217	2723884	226141	3	0	9
Heterogeneous linear controls	14725548	13769217	2723884	226141	3	24	78
Nonlinear homogeneous controls	14725548	13769217	2723884	226141	3	0	59
Full model	14725548	13769217	2723884	226141	3	24	1324

Note: For each panel and rung of the empirical ladder, the table reports the filtered sample size, the size of the largest connected component, the numbers of workers and firms, the number of continuous covariates, the number of interaction groups, and the number of control columns actually partialled out before the KSS decomposition.